21IPC501J
Research Methodology and Publication Ethics

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Module-1 – RESEARCH PREPARATION, PLANNING AND RESOURCES

Topics

- Research – objectives, definitions – knowledge economy;
- Perspectives of stakeholders;
- Types of research;
- Critical & creative thinking;
- Generating research topic;
- Identifying broad research question and objectives – design of questionnaire;
- Characteristics of a good research – research design;
- Methods of scientific enquiry – theoretical, experimental and empirical;
- Hypothesis – formulation & testing.

Module-1 – RESEARCH PREPARATION, PLANNING AND RESOURCES

Topics

- Sources of information – ICT enabled tools;
- Databases, repositories, public and private sources, indexes;
- Literature search – keyword;
- Backward and forward search;
- Quality measurement tools: journal metrics, author-level metrics;
- Meta-analysis of research findings;
- Literature review: grouping, analysing & comparison: Identification of research Methods;
- Intellectual Property Rights;
- Copyright – patent –geographical indication – industrial designs – technology transfer.

DEFINITIONS OF RESEARCH

Research :

Research = Re + Search

- 'Re' means again and again and 'search' means to find out something.
- The word research is combination of "re" and "search," which means a systematic investigation to gain new knowledge from already existing facts.
- It is basically search for truth.
- The following is the process,

DEFINITIONS OF RESEARCH

- Research is a systematic process where a person repeatedly observes a phenomenon, collects and analyzes data, and then draws conclusions.
- Example:
  - A student notices that some plants in the garden grow faster than others.
  - They observe this repeatedly.
  - They collect data on sunlight, water, and soil type.
  - They analyze the data to find the relation.
  - They reach the conclusion that plants with more sunlight grow faster.
  - So, research : “Carefully observing, collecting data, analyzing it, and concluding to understand or solve a problem.”

DEFINITIONS OF RESEARCH

Research :

- Research contributes significantly to the progress of the nation as well as an individual with commercial, social, and educational advantages.
- Research is an important parameter to judge the development of any nation.
- Research is an important component of private and government sectors.
- Nowadays, interdisciplinary research is at high demand.
- Research should always aim at providing efficient solutions to routine problems.
- Researchers should carefully choose the appropriate research method and follow a research process by referring to existing theories.

Definition of Research in Common Parlance

Research has many definitions in common parlance:

- Search for knowledge
- Scientific and systematic search for pertinent information on a topic
- Systematic investigation of inquiry specifically searching for new facts in any discipline
- Movement using known facts to unknown or unexplained issues
- Art of scientific investigation
- Voyage of discovery
- Events primed by inquisitiveness as driving force
- Academic activity used in a technical sense

DEFINITIONS OF RESEARCH

- Research refers to the systematic method consisting of enunciating the problem, formulating a hypothesis, collecting the facts or data, analyzing the fact and reaching certain conclusions either in the form solutions(s) towards the concerned problem or in certain generalizations for some theoretical formulation.

STAGES OF RESEARCH

- Selection of a Research Topic
- Definition of a Research Problem
- Literature Survey and Reference Collection
- Assessment of Current status of the topic chosen
- Formulation of hypothesis
- Materials and Methods (Research Design)
- Conduct of Experiments (Actual Investigations)
- Data Analysis
- Interpretation of the Results
- Findings and conclusions

Technical Definitions of Research

- Defining and redefining problems, formulating hypothesis or suggested solutions, collecting, organizing and evaluating data, making deductions and reaching conclusions, and at last carefully testing the conclusions to determine whether they fit the formulating hypothesis. (Clifford Woody)
- The manipulation of things, concepts or symbols for the purpose of generalizing to extend, correct or verify knowledge, whether the knowledge extends to the construction of a theory or the practice of an art. (Slesinger and Stephenson)
- Research is a tool that is building a block and a sustaining pillar of every discipline — scientific or otherwise that one knows.
- Research refers to the systematic method consisting of enunciating the problem, formulating a hypothesis, collecting the facts or data, analyzing the fact and reaching certain conclusions either in the form solutions(s) towards the concerned problem or in certain generalizations for some theoretical formulation.

OBJECTIVES OF RESEARCH

Objectives of Research:

- The main purpose of research is to find solutions to unsolved problems using scientific procedures and also to understand various phenomena scientifically.
- In addition, one of the major objectives of research is to find out a hidden, undiscovered truth.
- There are various objectives behind undertaking research by individuals as well as various organizations.

Objectives of Research

- The purpose of research is to discover answers to questions through the application of scientific procedures. The main aim of research is to find out the truth which is hidden and which has not been discovered as yet. The objectives may be classified as:
- To gain familiarity with a phenomenon or to achieve new insights into it (studies with this object in view are termed as exploratory or formulative research studies)
- To portray accurately the characteristics of a particular individual, situation or a group (studies with this object in view are known as descriptive research studies)
- To determine the frequency with which something occurs or with which it is associated with something else (studies with this object in view are known as diagnostic research studies)
- To test a hypothesis of a causal relationship between variables (such studies are known as hypothesis-testing research studies)

Research and Knowledge Economy

- A poorly developed country is agriculture and manufacturing - based economy
- A developing country is manufacturing and service - based economy
- A developed country is knowledge - based economy
- Knowledge economy is a system of consumption and production based on intellectual capital, which refers to the ability to capitalize on discoveries emanating from basic and applied research
- Hence, a significant contribution to the knowledge - economy emanates from intangible assets such as research, technical support, and consulting and intellectual property rights and royalties
- Hence, research plays an important part in the growth of an economy

Research and Stakeholders Perspectives

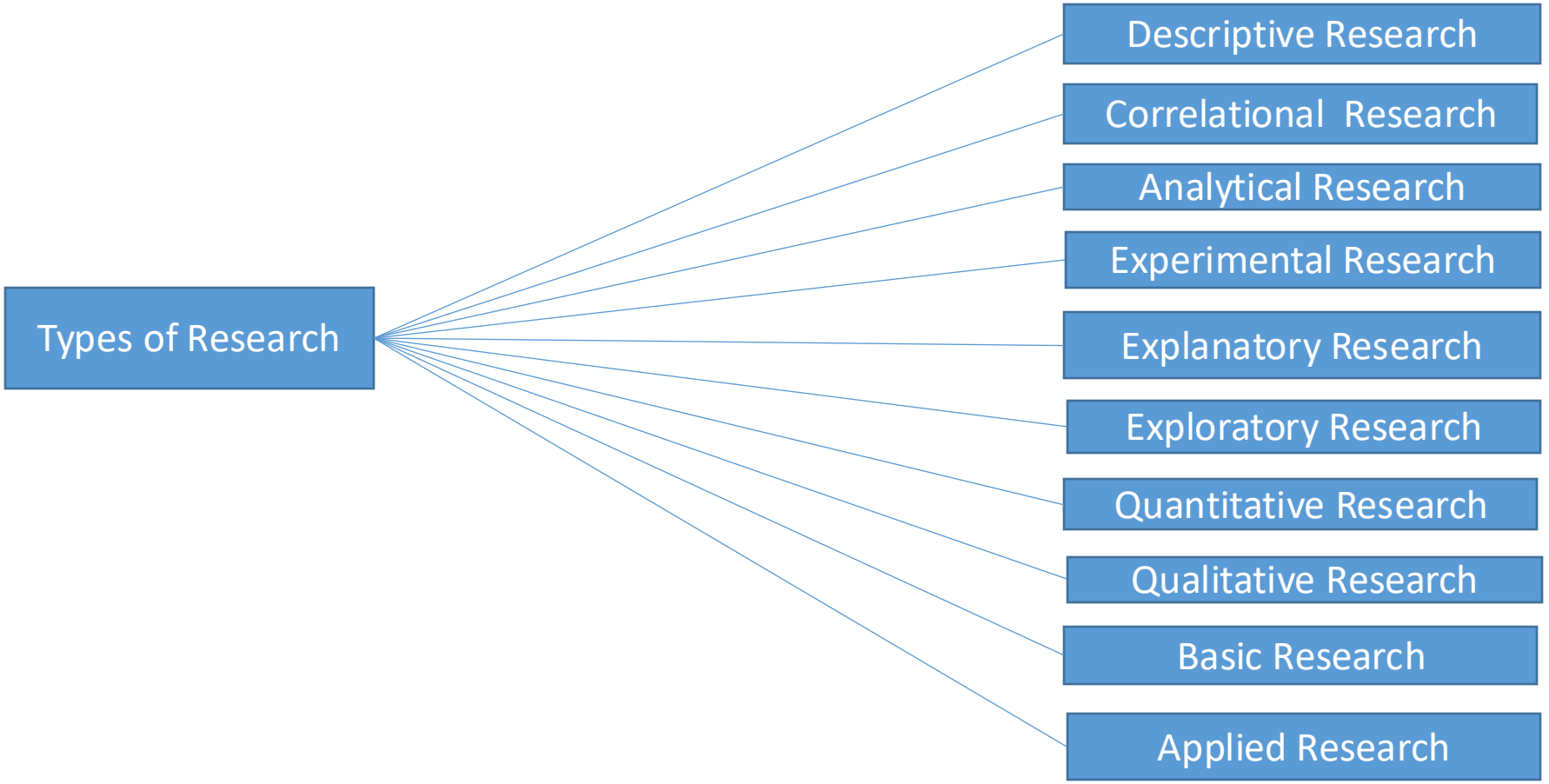
- Stakeholders are individuals or organizations who have an interest in your research project, or who affect or are affected by its outcomes.

Research and Stakeholders Perspectives

- They maybe groups and organizations, such as funders, donors, community members, advocacy organizations, patients, trade associations, businesses, policymakers, policy advocates (NGO), and product developers.
- A stakeholder may serve multiple roles and bring several perspectives to a project: setting priorities, shaping the research questions, designing or Implementing studies, funding programs, assessing applications, communicating findings, and moving the results into practice.
- Importantly. stakeholder engagement can bring knowledge and expertise different experiences and perspectives to your research, provide data and resources, build shared understanding, and make information from your research relevant for effective decision making.
- Hence stakeholders perspectives play a significant role in research

Types of Research

- Types of research may be categorized depending on applicability, the mode of enquiry in conducting the study, and major objectives of the study. In fact, a research problem may use multiple research types.



Ref : Research Methodology A Practical and Scientific Approach, Biragi and Munot, CRC Press, 2019

TYPES OF RESEARCH AND ITS GOALS

Pure Research (or) Basic (or) Fundamental (or) Theoretical Research:

- There is no immediate need, but new theories can be added to the knowledge cluster.
- This type of research may solve problems but may not have practical applications.
- It has a broader scope compared to applied research.
- Research concerning some natural phenomenon or relating to pure mathematics are examples.

Example: Newton's laws of motion.

- This has been applied in many product design and testing.

Applied Research:

- Applied Research tries to solve an immediate specific problem faced by industry or society.
- The obtained solution can be deployed to solve the problem.
- The duration of applied research is shorter as a quick solution is expected.
- The outcome of the applied research either address the unsolved problem or improve the existing solution.

Example:

- 1. An optimized search problem on the Internet is an example of applied research in the computer engineering domain.

Example:

- 2. "Analysis of cell/body organ behavior in cancer" is an example of hybrid research.
- A researcher may use data analytics, image processing, algorithms, and knowledge of the medical industry.

Distinguish between pure research and applied research?

Pure research enriching the theory whereas applied research enriches the application of the theory.

Types of Research

Descriptive Research — *Ex post facto* research

- To portray accurately (survey) the characteristics of a particular individual, situation or a group.
- Mostly involves primary data — the researcher has no control over the variable — he can only report what has happened or what is happening. *(Ex post factor)*
- Some examples are:

❖Number of man-days lost in a factory because of accidents

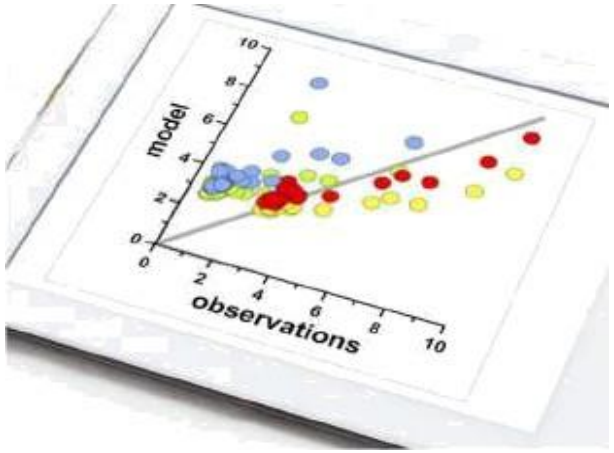
❖Number of COVID related deaths due to the pandemic

❖Gender, age and other demographics of a population

DESCRIPTIVE RESEARCH

Types of Research

Correlation Research — *constructing models*



- To establish the existence of relationships between two or more aspects of a situation
- It explores the relationship or association between incidences, variables etc.
- Examples include:
  - ❖Effect of cigarette smoking on incidences of lung cancer
  - ❖Effect of modern lifestyle on obesity
  - ❖Effect of leaf length on leaf weight

Types of Research

Analytical Research - *cause and effect*

- Analytical research focuses on cause and effect.
- It tells us why our certain hypothesis is true or false, which requires critical thinking skills and careful assessment of the facts.
- Examples are:
  - ❖Why has the value of INR slid down against the US Dollar over the last three decades.
  - ❖What is the cause for the high rate of suicides in a highly educated society

Types of Research

Experimental Research

- It is a study that adheres to a scientific research design.
- It includes a hypothesis, a dependent variables that can be manipulated by the researcher, and independent variables that can be measured, calculated and compared.
- The researcher collects data and results will either support or reject the hypothesis.
- Example of this is the type of research where investigator optimizes the desired independent variable by manipulating the dependent variables.

Types of Research

Explanatory Research

- It is also called causal research and tries to analyse and justify the reason behind the occurrence of particular phenomenon or association between the variables.
- It basically answers the “Why” type of questions.
- It aims to explain why a relationship, association, or interdependence exist.
- Example is: Why do customers prefer to purchase a certain colour packaged items more than others

Types of Research

Exploratory or Formulative Research —

EXPLORATORY RESEARCH

- *Explore new frontiers*
- To gain familiarity with a phenomenon or to achieve new insights into it.
- It does not aim to provide a conclusive answer to the research questions, but merely explores the research topic to help develop a hypotheses (rather than its testing).
- Example is a product launch in a particular city to judge the customer response before it can be launched nation-wide.

Types of Research

Quantitative Research

- As the name suggests, this involves measurements of quantities of a particular characteristic that can be used for the research study.
- It uses statistical analysis on parameter values to reach conclusions.
- Example is the number of fatalities of the COVID pandemic with respect to different criteria such as rural, urban residents, region-wise, educational qualification etc.

Qualitative

Quantitative

Types of Research

Qualitative Research

- This research is about “what” people think and “why” they think so.

  So, it deals with human behaviour
- So it is empirical research where data are not in the form of numbers.
- It is based on the disciplines of social sciences like psychology, sociology, and anthropology.
- Example is the study of the behavioural response of employees to the

  changes occurring in an organization.

Qualitative Research

Types of Research

- Basic Research is pure or fundamental research.
- It aims to obtain a better understanding of a subject, phenomenon or basic law of nature.
- This type of research is primarily focused on the advancement of knowledge rather than solving a specific problem.
- Theories in basic sciences and mathematics are examples of basic research.

Types of Research

Applied Research

- Applied research tries to solve an immediate specific problem faced by society.
- The obtained solution can be deployed to solve the problem.
- The outcome of applied research should either address the unsolved problem or improve the existing solution.

- Example of applied research is the discovery of drugs to treat diseases

Types of Research

Some other types of research are:

- **Diagnostic Research** — to determine the frequency with which something occurs or with which it is associated with something else (Health Science).
- **Hypothesis-Testing Research** — to test a hypothesis of a causal relationship between variables by forming suitable hypotheses (model assisted studies).
- **Overt and Covert Research** — overt research means that everyone knows who the researcher is and what research is being conducted. Covert research means that it is an under-cover work and nobody knows who the researcher is and what researcher is being conducted.

Classification of Research

Basic Research

Enriching the theory

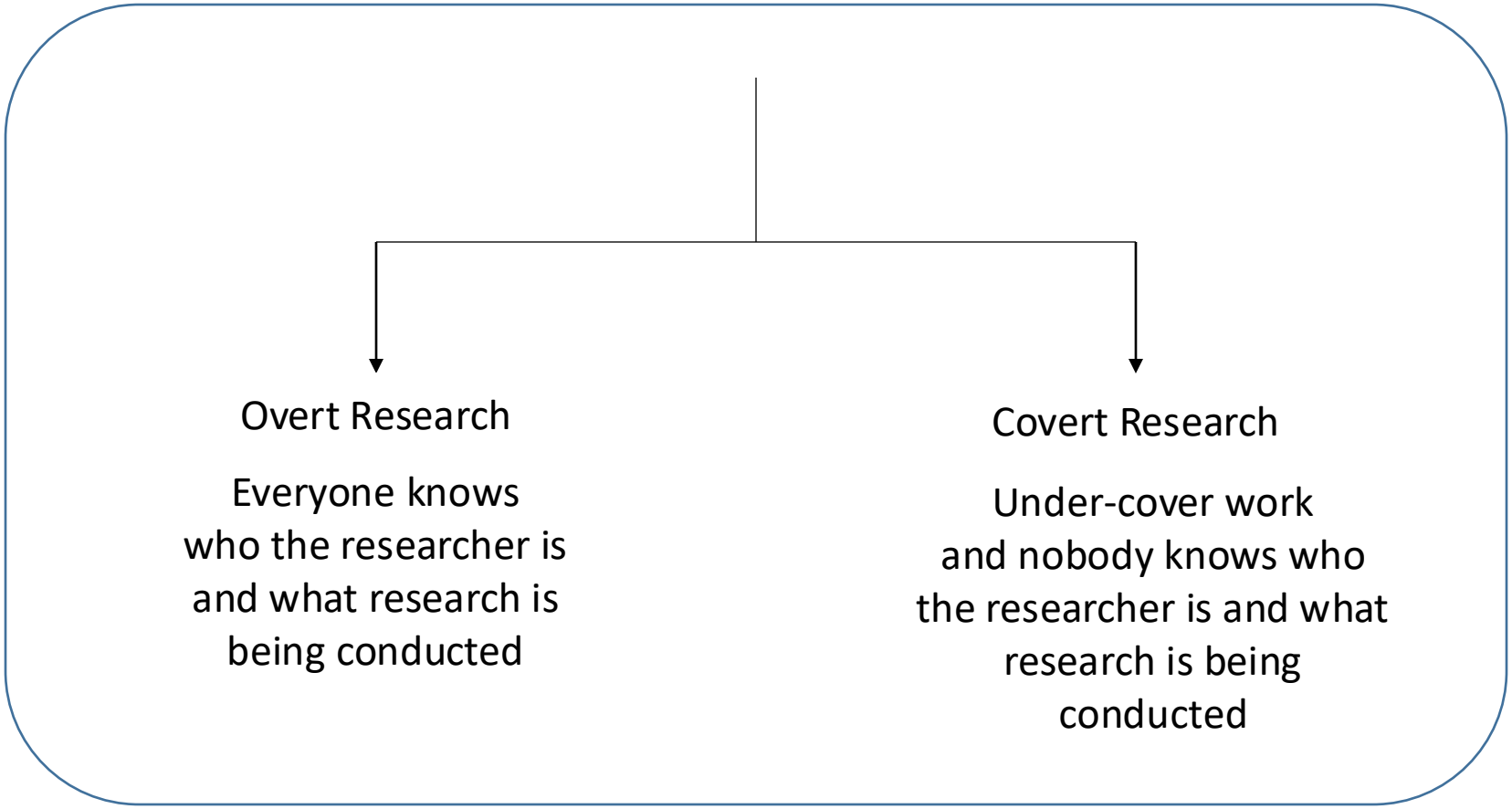
Applied Research

Enriching the applications of theoretical knowledge

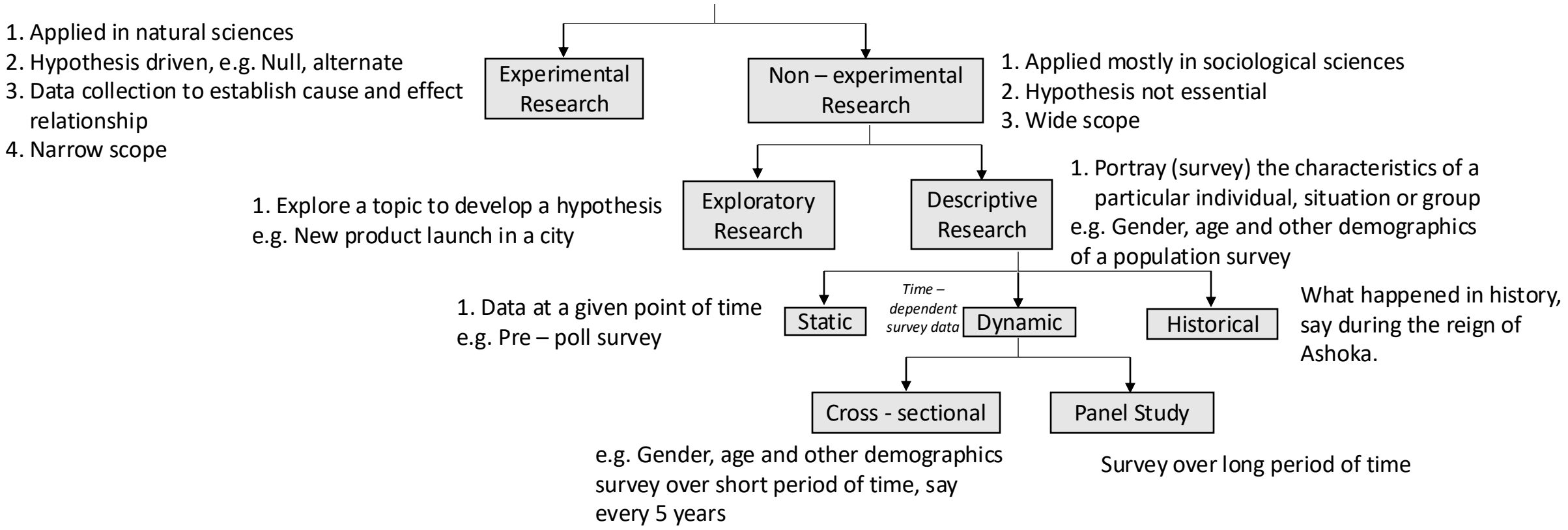
Classification of Research

Basic research	Applied Research
It more theoretical also explores the information that have not applied yet	Applied research is more practical and descriptive in nature
The scope of basic research is often universal at can be applied in diverse concepts	The scope of applied research is largely focused on the particular topic to get the answer
It is lesser involved in improvement of technology	It is comparatively highly applied to improve the technology and covers the larger areas
It aims to predict future research and analyze the gap between the researches prior done	Applied seeking to prevent the predicted problem having potential to create challenges in the future
It is driven by curiosity	It is usually driven by clients and conducted to understand the fundamental concept required to solve the peoples problem
In case of basic research, it is primarily applicable to academic areas	It is closely linked to commercial products
Basic research occurs in highly sterile and controlled environment	It takes place in world setting where other unexpected variable may intervene
Example - Research on identifying secondary symptoms of human papilloma virus	Example- Research into marijuana healing properties
It approaches flexible time scales where choices of the topic is determined by the researchers	It approaches tight time scale where objectives are negotiated with originator

Classification of Research



Classification of Research



Research Methods vs. Research Methodology

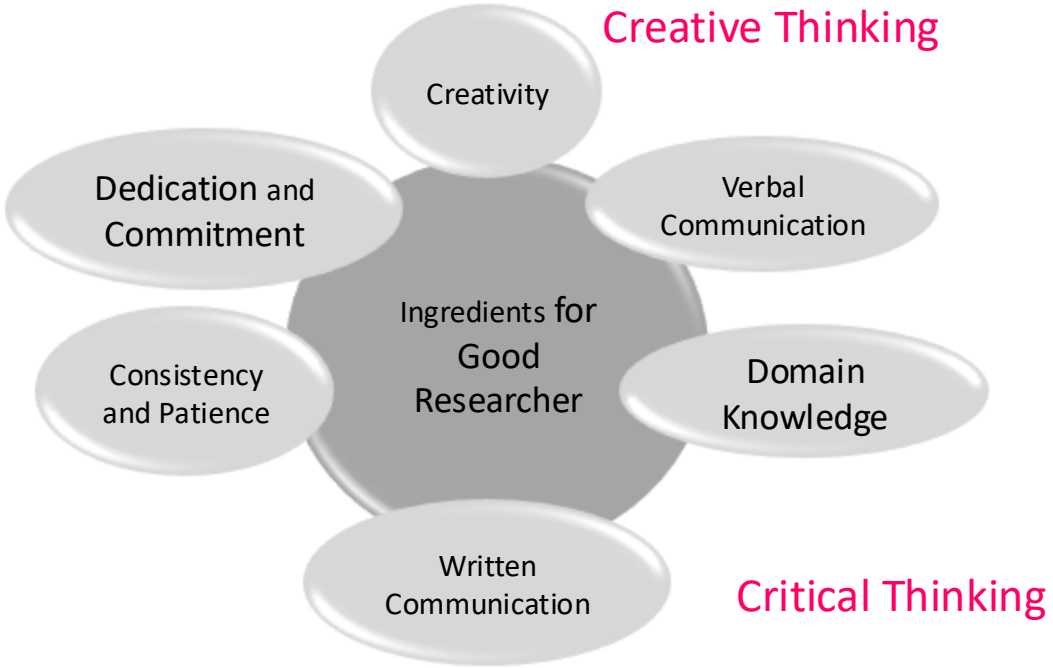
- **Research methods** may be understood as all those **methods/ techniques** that are used for conduction of research.
- Since the object of research, particularly the applied research, is to arrive at a solution for a given problem, the available data and the unknown aspects of the problem have to be related to each other to make a solution possible.
- Keeping this in view, research methods can be put into the following three groups:
    1. In the first group we include those methods which are concerned with the collection of data, *e.g.* **experimental (instrumental) methods, survey** etc.
    2. The second group consists of those **statistical techniques** which are used for establishing relationships between the data and the unknowns
    3. The third group consists of those methods which are used to evaluate the accuracy of the results obtained

Research Methods vs. Research Methodology

- Research methodology is a way to systematically solve the research problem in a logical step-by-step manner.
- Hence, It is necessary for the researcher to know not only the research methods / techniques but also the methodology.
- The scope of research methodology is wider than that of research methods.
- Thus, when we talk of research methodology we not only talk of the research methods but also consider the logic behind the methods we use in the context of our research study
- Explain why we are using a particular method or technique and why we are not using others so that research results are capable of being evaluated either by the researcher himself or by others.

Attributes of Good Researcher
Role of Critical Thinking and Creative Thinking

- A good researcher should possess technical as well as professional skill-sets
- The technical skill-set deals with domain knowledge and requires critical thinking
- The professional skill – sets deal with the other aspects, including creative thinking mentioned in the figure
- Both critical and creative thinking are essential for research



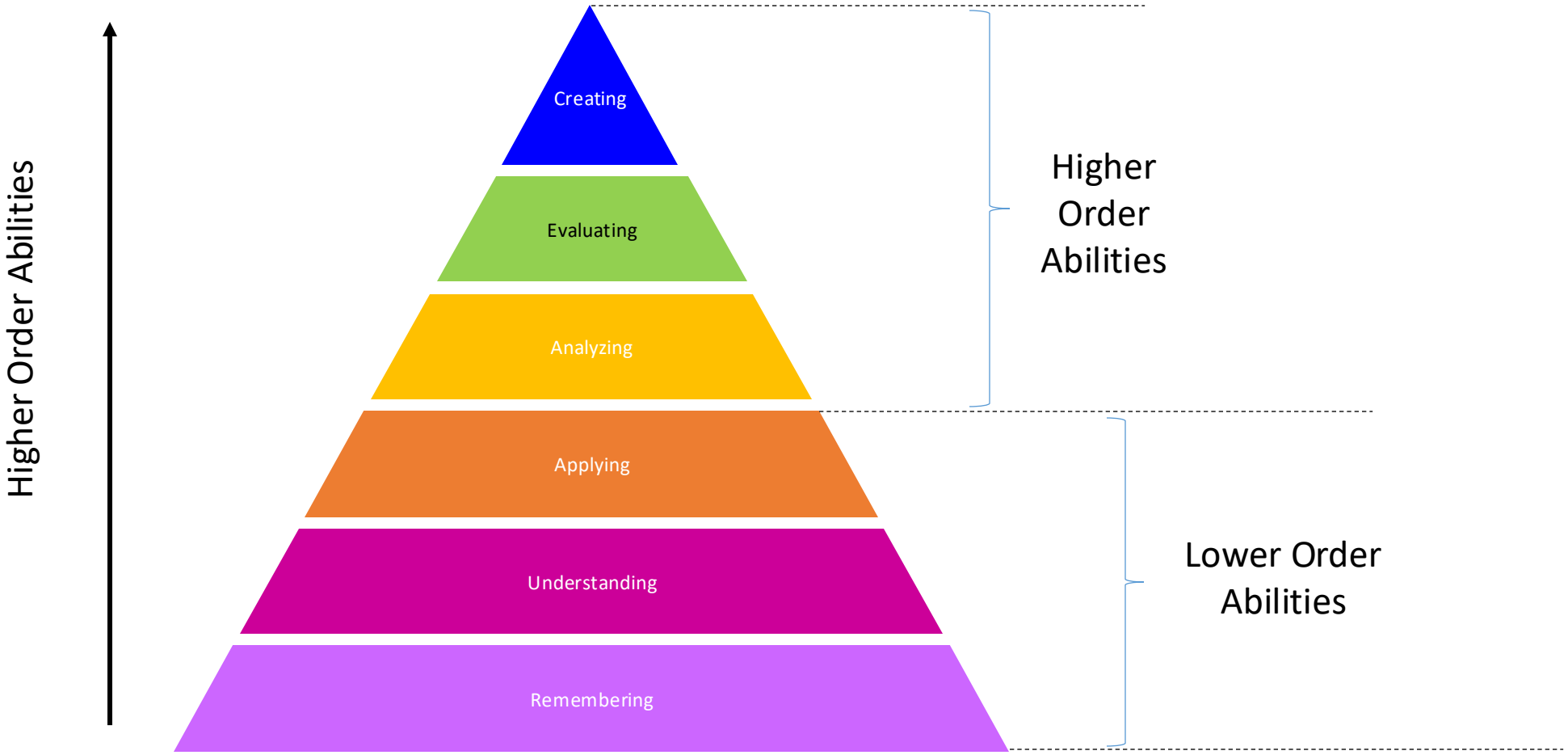
Critical Thinking *vs.* Creative Thinking

- Critical thinking is the ability to clearly and logically evaluate information that is presented. So, it is an analytical action.
- It can also be defined as the purposeful and reflective judgment about what to believe and what to do in response to the observations, experience, verbal or written expressions or arguments.
- Creative thinking is about reflecting on different perspectives or truths and then generating new, novel, or useful ideas. So, it is an generative action.
- Critical and Creative thinking are complementary skills which are used in different stages when trying to solve a problem or forming a judgment about something.

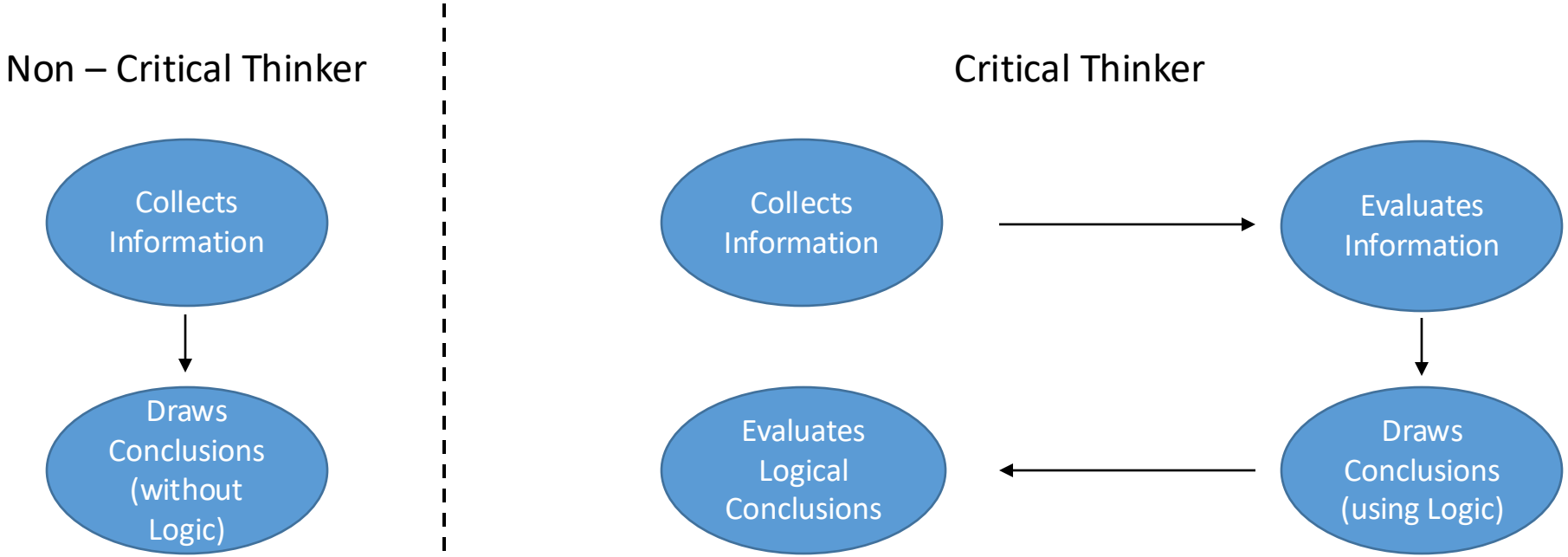
Critical Thinking and Bloom's Taxonomy

- Critical thinking is the ability to clearly and logically evaluate information that is presented to us. So, it is an analytical action.

Bloom’s Taxonomy Framework



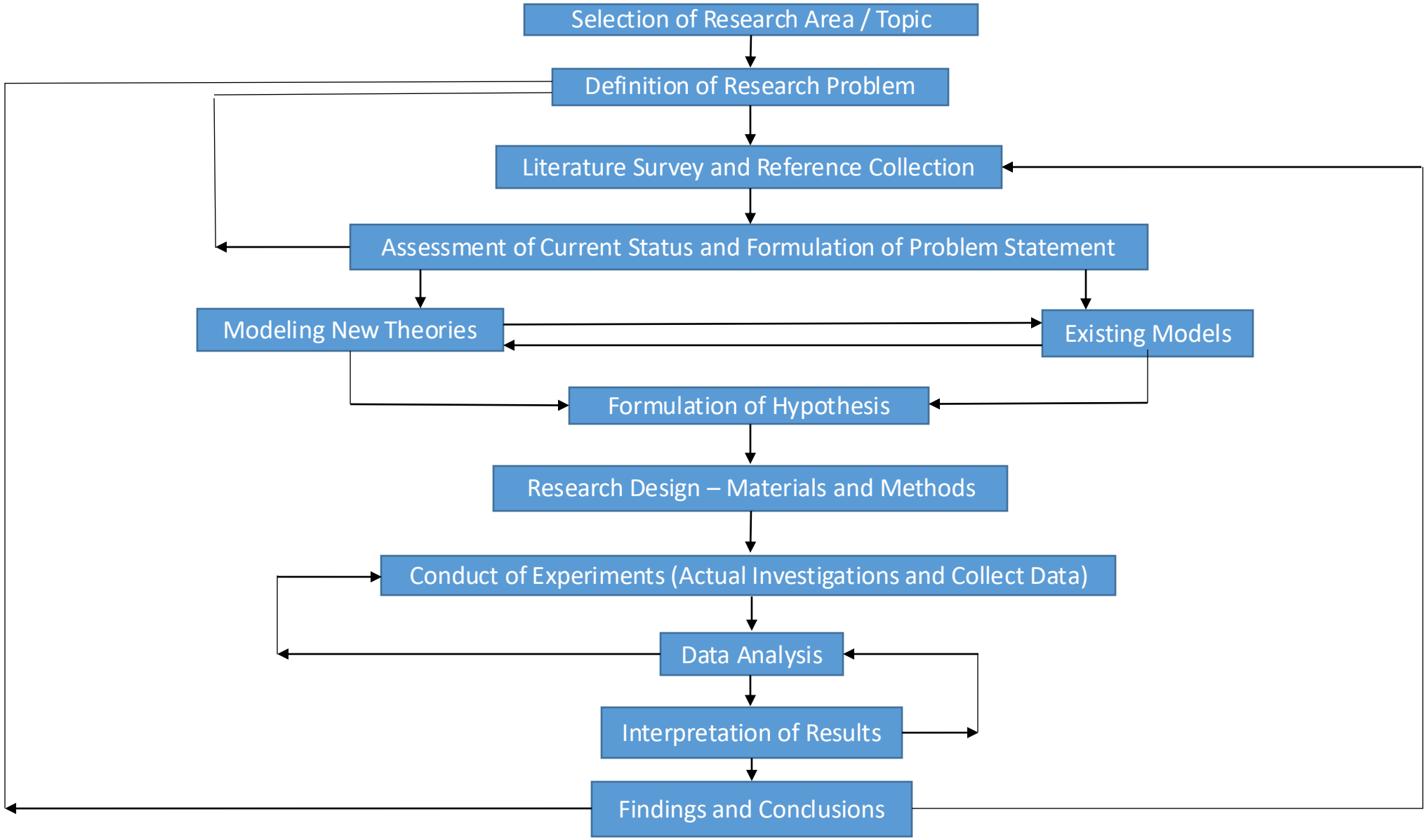
Tools of Critical Thinking



There are four stages in critical thinking:

i. Collecting data
ii. Evaluating it
iii. Drawing logical conclusions and
iv. Evaluating the validity of the conclusion

Stages of Research



Research Problem

Identification of Research Area and Topic

- Choice of Research Area of the problem maybe motivated by:
  - Interest of the researcher due to self or other close associates
  - Previous domain knowledge of the researcher
  - Results of a critical literature survey of the donna in/area of interest
- A guide can supervise, but it is the student who has to decide on the research area, topic and particular problem
- Important points to be taken into account
  - Over explored research areas to be avoided
  - Controversial research areas to be avoided
  - Very narrow or too vague problems to be avoided
  - Familiar and feasible subjects (for guide) to be chosen
  - Research significance, researcher background, cost estimate and time factor to be considered
  - Preliminary study needed before deciding on the work

Research Problem

Identification of Research Area and Topic

- Choice of Research Topic is identified based on combination of
  - Specialization
  - Domain knowledge
  - Interesting or challenging topic
  - Doable within a specific time frame
  - Guidance of supervisor
  - Discussion with peers and experts
  - Critical literature survey to identify gaps of knowledge

Research Problem

Originality (Novelty)of Research Area and Topic

The originality can be confined to:

- New interpretation to support known facts

  e.g. COVID infection using known spike protein amino acid sequence

- Unearthing new evidences to support known facts

  e.g. Mutant COVID infection using new spike protein amino acid sequence

- Proposing new concept to explain a known phenomena

  e.g. cellular signaling mechanism involving progression of a disease, say cancer metastasis

- Proposing a new model to explain a known phenomena

  e.g. mathematical model predicting the descent of a rover onto the surface of the Moon

Research Problem

Originality (Novelty)of Research Area and Topic

- Utilization of a novel technique in an untested area

  e.g. mass spectrometry based secretomics for cancer biomarker discovery

- Utilization of a novel methodology to solve a known problem

  e.g. controlled real-time response based release of insulin into the blood stream to maintain *sugar level*

- Empirical work that is not repeat of an earlier work
- Furthering the work of some earlier researcher as suggested in his/her thesis or publication

Creative and critical thinning should foster cross-breeding of ideas and applications in different contexts to solve problems.

Research Problem

Approach for Selection of Research Area and Topic

There is no universal approach for this! One approach is:

- A direction or research interest area is chosen based on the background of the researcher
- A set of keywords is compiled for searching of literature
- A set of recent and well cited review articles, say 15 to 20 is chosen that is related closely to what was chosen as area by the researcher
- The keywords are refined further to identify related recent as well as distant past research publications. Cross referencing may be used for literature search.
- The emphasis (problem addressed, challenges overcome, solution proposed, novelty in methodology, conclusions *etc.)* of each article is noted down

Research Problem

Approach for Selection of Research Area and Topic

- The emphasis of the article in following terms is noted:
  - Problem addressed
  - Approach methodology employed — metrics, evaluation tools, data interpretation, result analysis
  - Challenges overcome
  - Solutions proposed
  - Significance of conclusions
- Based on this some questions can be asked:
  - Is same approach methods and methodology used in all articles and can be it be replaced with better ones
  - If so, what are the advantages and disadvantages *etc.*
  - Is further exploration possible in the research area *etc.*

Research Problem

Approach for Selection of Research Area and Topic

Other approaches are:

- Research focus areas of funding agencies such as DST, DBT, ICMR, DRDO etc. are:
  - Cleantech - clean technology for fossil fuels
  - Carbon capture and storage — environmental aspects
  - Biorefinery — efficient utilization of natural resources
  - Defense and Aerospace — space research
  - Food and Agriculture — nutrition and self sufficiency
  - Electric and hybrid vehicles — smart vehicles and alternate fuels
  - Electronic Systems and Manufacturing - Semi-conductor tech.
  - New and renewable energy - solar, wind *etc.*
  - Advanced and Smart Manufacturing
  - Biotechnology - Vaccine production, recombinant tech. etc.
  - Biomedical research — fundamental and applied disease research, equipment's

Research Problem

Approach for Selection of Research Area and Topic

Other non-regular approaches are (contd.):

- Industrial R & D activities and its budgets as per annual reports of companies
- Latest edition text books of emerging areas
- Further scope of research as mentioned in PhD theses
- Discussion with experts
- Attending seminars, conferences to listen to experts

Based on all these, a one or two-page summary of the research area, specific topic of interest and its current status can be documented. This can be used to the formulate the problem statement.

Research Problem Statement

Introduction

- The first step in solving any research problem lies in first clearly identifying and stating the nature of the problem.
- Such a statement is called as a "Problem Statement" and hence can be defined as a clear description of a problem that needs to be solved.
- This simple, yet effective problem-solving technique (which forms part of the design thinking process) requires breaking down the problem by asking questions such as "who does it affect?" and "how does the problem make the stakeholders feel?" *etc.,* so the issue maybe stated in a concise manner and quickly and easily understood by all stakeholders.

Problem Statement Formulation

Problem Definition

"A problem well defined is a problem half- solved"

- Defining a problem involves laying down boundaries within which a researcher shall study the problem with a predetermined objective.
- It is important to develop a well developed problem definition early in the research to focus on the deliverables of the specific objectives.
- Answer to questions such as the ones that follow will help in this task:
  - What are the data to be collected?
  - What are the relevant characteristics of this data?
  - What are the relations that need to be explored?
  - What are the methods, techniques etc. to be used?

Problem Statement Formulation
Suggested Techniques / Steps

The following 5 steps maybe utilized for the formulation

1. General Statement of the Problem The research problem is stated in a general and broad manner keeping in mind the probable strategy for the solution
2. Understanding the Problem Origin and nature of the problem can be understood by reading news papers and magazines, general literature etc. The research problem could be discussed with guide and fellow
3. Surveying Literature A broad reading of available literature provides an idea about the shortcomings, difficulties, challenges and probable analytical solutions to the issues concerning the research area. This helps to narrow down the topic.

Problem Statement Formulation
Suggested Techniques / Steps

4. Develop Ideas through Discussion Discussion with fellow researchers, guide, experts etc. will help to narrow down to the particular research problem and topic as well as probable solutions.
5. Rephrasing the Problem Statement The research problem can be formulated into a working proposition. This maybe an iterative process starting again with the first “General Statement” step.

Finally, the problem statement should also take care to explain hypothesis, technical terms, assumptions made, criteria for selection of research problem area and topic, source of data to be acquired, type of data analysis, scope and limitations if any and anticipated solution *etc.*

Questionnaire

- A questionnaire is a research instrument consisting of a series of questions (or other types of prompts) for the purpose of gathering information from respondents through survey or statistical study.

- A questionnaire is a formalized set of questions to generate the data necessary for a accomplishing a research project's objectives.
- In general, questions should flow logically from one level to the next higher level of complexity.

Designing a Questionnaire
Ten Points to be Considered

1. Must be initially related to final objectives of the investigations
    - Specify the information needed
2. Questions must follow logical sequences
    - Arrange the questions in proper order
3. Specify the type of interviewing method
    - Biased questions should be avoided
4. Proper structuring of the question is must
    - Decide on the question structure
5. Should be brief, clear and simple
    - Determine the content of individual questions
    - Determine the question wording

Designing a Questionnaire
Ten Points to be Considered

6. Reproduce the questionnaire
    - Question should have appropriate and to it
7. Eliminate bugs by pretesting
    - Conflict between interviewer and respondent should be avoided
8. Questions outside the respondents experience should not be asked
    - To overcome the respondent's unwillingness to answer
9. Questions should test the memory of the respondent
    - To overcome the respondent‘s inability to answer
10. Identify the form and layout
    - To gather factual data

Hypothesis
Introduction and Definition

- In common parlance, hypothesis simply states a mere assumption or some supposition to be proved or disproved.
- A hypothesis is written in such a way that it can be proven or disproven by valid reliable data. (Griennell & Stothers)
- A tentative statement about something, the validity of which is usually unknown. (Black & Champion)

- In other words, it can be said that a hypothesis is a hunch, assumption, suspicion, assertion or an idea about a phenomenon, relationship or situation, the reality or truth of which is not known.

Functions of an Hypothesis

A hypothesis though not an essential feature, brings clarity to the research with the following functions:

- The formulation of a hypothesis provides a study with focus. It tells what specific aspects of a research problem to investigate.
- A hypothesis tells what data to collect and what not to collect, thereby providing focus to the study.
- As it provides a focus, the construction of a hypothesis enhances objectivity in a study.
- A hypothesis may enable the researcher to add to the formulation of theory. It helps in concluding specifically what is true or what is false.

Characteristics of an Hypothesis

A hypothesis must possess following characteristics:

1. Hypothesis should be clear and precise. If the hypothesis is not clear and precise, the inferences drawn on its basis cannot be taken as reliable. In other words, it should be non contradictory one.
2. Hypothesis should be capable of being tested. In a swamp of untestable hypotheses, many a time the research programmes have bogged down.
3. Hypothesis should state relationship between variables, if it happens to be a relational hypothesis.
4. Hypothesis should be limited in scope and must be specific. It Should be remembered that narrower hypotheses are generally more testable

Characteristics of an Hypothesis

5. Hypothesis should be stated as far as possible in most simple terms so that the same is easily understandable by all concerned
6. Hypothesis should be consistent with most known facts i.e., it must be consistent with a substantial body of established facts.
7. Hypothesis should be amenable to testing within a reasonable time.
8. Hypothesis must explain the facts that gave rise to the need for explanation. This means that by using the hypothesis plus other known and accepted generalizations, one should be able to deduce the original problem condition.

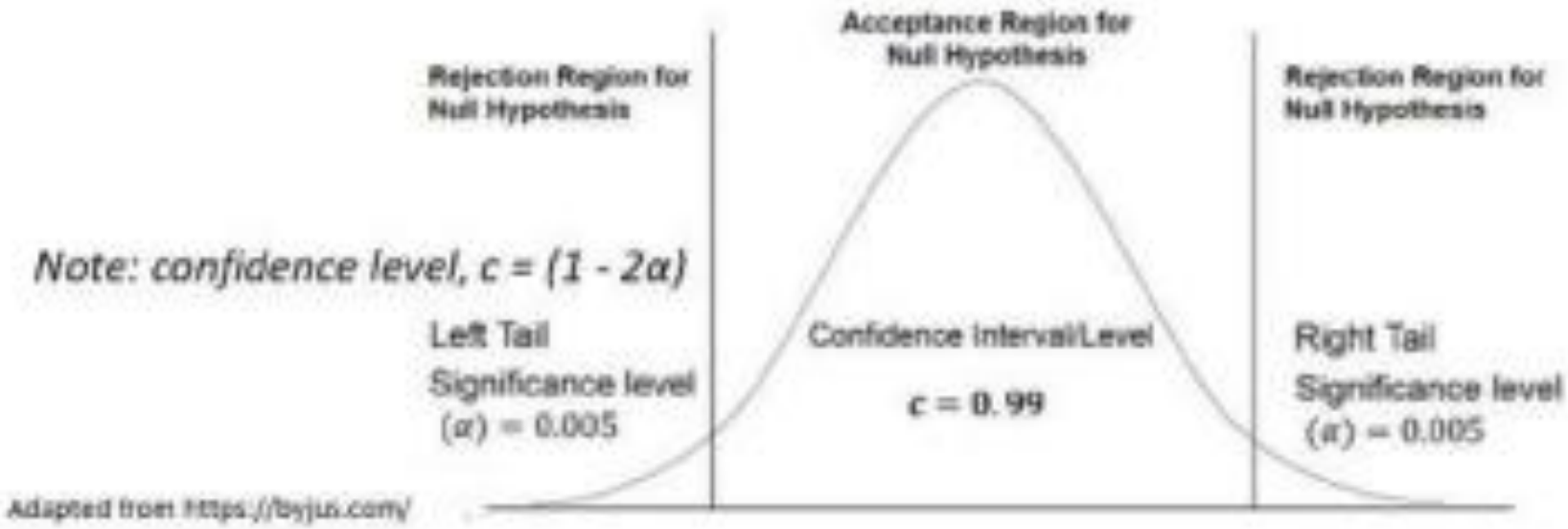
Types of Hypothesis

- Null Hypothesis: The null hypothesis is a commonly accepted fact. It's the default, or what we'd believe if the experiment was never conducted. It is deemed true until proven wrong.
- For eg., there is no significant difference between two groups, say *A.* and B.
- It is a hypothesis of no difference. Researchers work to nullify or

  disprove null hypotheses. It is symbolized as H_0.
- Alternative Hypothesis: This is the opposite of the null hypothesis, demonstrating or supporting a statistically significant result.
- For eg,, if we state that group A is superior or the group B is inferior, we are then stating the alternative hypothesis.
- By rejecting the null hypothesis, we accept the alternative hypothesis. It is denoted by H_1 or H_a.

Level of Statistical Significance (α)

- Level of significance (α): The level of significance is defined as the probability of wrong elimination of null hypothesis when in fact, it is true. It is the opposite of the confidence level (denoted by *c).* The c level is usually set ranging from 0.1 to 0.005 (10% to 0.5%).
- An α = 0.05 means 5% chance of finding alternate hypothesis to be true, thereby rejecting null hypothesis, when in actuality, null hypothesis is true.
- In the case of the effect of fertilizer on plant growth example, it is the probability that the null hypothesis of no effect of fertilizer on plant growth is rejected when it is indeed true.
- The success of the statistical test depends on the availability of correct and reliable data and researchers skill in interpreting the results

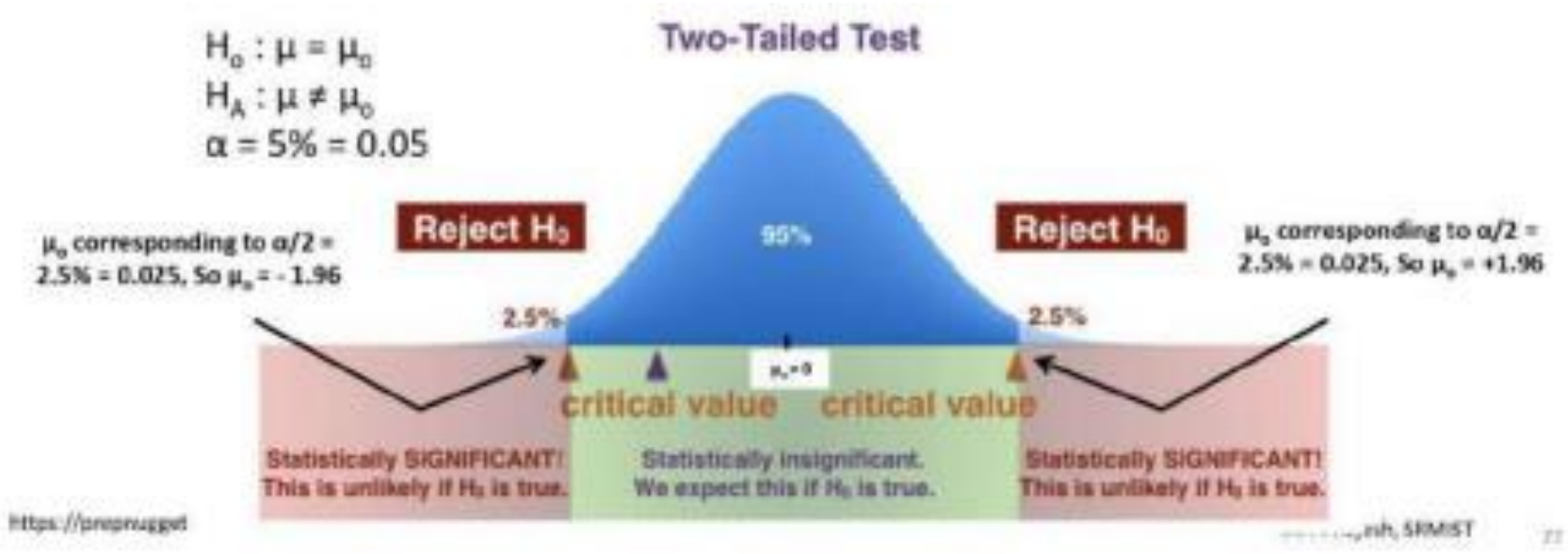


Two – Tailed and One – Tailed Test

The two-tailed test is applicable when a given H, may be rejected with the value of the statistic being either sufficiently small or sufficiently large.

$$H_1: \mu \neq \mu_0, \text{ because either } \mu > \mu_0 \text{ or } \mu < \mu_0$$

Hence there are two rejection regions. The H_0 is rejected if the sample mean falls in either one of the two-tail regions under the far left or far right of the distribution. Please note $\mu_0 = \pm 1.96$ for $\alpha = 0.05$ which maybe obtained from tables in the figure below.



Two – Tailed and One – Tailed Test

The one-tailed test is applicable when a given H_0 may be rejected with the value of the statistic being sufficiently large (in the second case it might be sufficiently small).

$$H_1: \mu \neq \mu_0, \text{ because } \mu > \mu_0 \text{ (in the second case } \mu < \mu_0)$$

Hence there is one rejection region. The H_0 is rejected if the sample mean falls in the single tail region under the far right (second case being far left of the distribution). Please note $\mu_0 = +1.645$ for $\alpha = 0.05$ which maybe obtained from tables in the figure below.

Hypothesis Testing Procedure

1. Set up the hypothesis (H_0 H_a)
2. Select the significance level, say α = 0.05 or 0.01 *etc.*
3. Identifying test statistic that can be used to test the null hypothesis (large sample is n > 30, and small sample is n < 30)
4. Obtain the observed/calculated value based on the data using the test statistic
5. Obtain the expected value (tabulated value) from the designated table
7. Inference:
8. If the calculated value of Z is less than tabulated value z_α then H_0 accepted.
9. If the calculated value of Z is greater than tabulated value z_α then H_0

- rejected.

Hypothesis Testing with Example

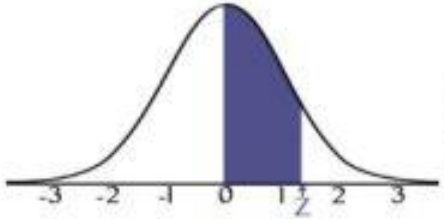
- A manufacturer of switches claims that data obtained over the years, show that their switches lasts for a mean of 2,000 hours with a standard deviation of 200 hours.
- Later, the R&D department claimed that they have came up with some innovation that might extend the life of the switches.
- To test this, the company conducted a test with 100 switches which showed that the they now had an mean life of 1,960 hours with a standard deviation of 180 hours.
- Based on these results, can the manufacturer claim that this has led to a change in the life span of the switches? Frame a hypothesis and test it with a significance level of 5%.

Hypothesis Testing with Example

- **Population Parameters:**
  - Population mean, μ = 2,000 hours of switch life
  - Population standard deviation, σ = 200 hours of switch life
- **Hypothesis:**
  - We can propose Null Hypothesis : $H_0 : \mu_0 = 2{,}000$
  - Alternate Hypothesis would then be : $H_1 : \mu \neq \mu_0$, So $\mu > \mu_0$, or $\mu < \mu_0$,
  - The significance level as suggested, α = 0.05
- **Sample Data :**
  - Sample size as provided, n = 100
  - Sample Mean, X = 1,960 hours of switch life
  - Sample Standard Deviation, $\sigma = 180$
- As n > 30, we can assume normal distribution which would make the standard deviation to be, $\sigma = 200$
- As the alternate hypothesis, $H_1 : \mu \neq \mu_0$, So $\mu > \mu_0$, or $\mu < \mu_0$, we can use the two-tailed test for testing the hypothesis.

Hypothesis Testing with Example

- The calculated value of Z would be given by the expression
  - $z = \frac{x-\mu}{\sigma/\sqrt{n}}$, Where σ is the sample standard deviation
  - Substituting $z = \frac{1960-2000}{180/\sqrt{100}} = -2.22$
- **Compare with Critical Z**
- **Calculated Z = -2.22**
- Given α = 0.05 , (1- α /2) =(1 - 0.05/2)=0.975 in the middle of the normal distribution
- So, to find the z-value that marks the boundary of the upper 2.5% tail, we look for the cumulative area up to that point:
- As -2.22 < -1.96, the z value falls in the tail region an therefore the **null hypothesis is rejected!**
- Hence the changes have indeed significantly changed the life of the switches!



STANDARD NORMAL TABLE (Z)

Entries in the table give the area under the curve between the mean and z standard deviations above the mean. For example, for $z = 1.25$ the area under the curve between the mean (0) and z is 0.3944.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998

Standard Normal Z Table

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

Example 1: Exam Scores
A teacher claims that the average exam score in her class is 75. A random sample of 36 students has an average score of 72, and the standard deviation is 6.Test the claim at 5% level of significance.

Example 2: Daily Water Consumption
It is believed that people drink on average 2.5 liters of water per day. A health survey of 30 people shows an average of 2.2 liters, with a standard deviation of 0.5 liters. Test if the actual average water consumption is different, at 1% significance level.

Example 3: A city claims that the average commute time to work is 30 minutes. A researcher takes a random sample of 25 workers and finds a sample mean commute time of 29.5 minutes, with a standard deviation of 3 minutes. Test whether the average commute time is different from the city's claim at a 5% significance level.

Example 1: Exam Scores

A teacher claims that the average exam score in her class is 75. A random sample of 36 students has an average score of 72, and the standard deviation is 6.Test the claim at 5% level of significance.

Since the test statistic z = -3 is less than -1.96

Example 2: Daily Water Consumption

It is believed that people drink on average 2.5 liters of water per day. A health survey of 30 people shows an average of 2.2 liters, with a standard deviation of 0.5 liters. Test if the actual average water consumption is different, at 1% significance level.

Since the test statistic z ≈ -3.29 is less than -2.576, it falls in the rejection region.

Example 3: A city claims that the average commute time to work is 30 minutes. A researcher takes a random sample of 25 workers and finds a sample mean commute time of 29.5 minutes, with a standard deviation of 3 minutes. Test whether the average commute time is different from the city's claim at a 5% significance level.

Since the test statistic z = -0.833 is between -1.96 and 1.96, it falls in the accepted.

Literature
Definition and General Characteristics

- A body of work that has been distributed, written or published on a given topic or concept
- Can be in multiple formats, depending on the academic subject or discipline
- Various intents: inform, support, persuade, update, or dispute

450 Year Old Banyan Tree in Adyar, Chennai
Ref.: https://www.flickr.com/photos/69911601@N08/7864827350

General Characteristics of Literature

The literature consists of the following :

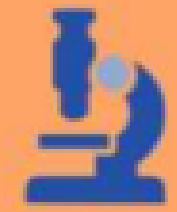
- Major works : key or seminal publications on a topic
- Other works or publications that build upon or respond to the major works.
- The literature is a continuously evolving network of scholarly works that interact with each other.
- The banyan tree in the adjoining picture is an analogy for the growing body literature that enhances our knowledge of the Universe

450 Year Old Banyan Tree in Adyar, Chennai
Ref.: https://www.flickr.com/photos/69911601@N08/7864827350

Types and Sources of Literature

- Literature falls into three categories : Primary, Secondary and Tertiary Sources

PRIMARY LITERATURE



- Original research and / or new scientific discoveries
- Immediate results of research activities
- Often includes analysis of data collected in the field or laboratory

EXAMPLES

- Original research published as articles in peer – reviewed journals
- Dissertations
- Technical reports
- Conference proceedings

SECONDARY LITERATURE

- Summarizes and synthesizes primary literature
- Usually broader and less current than primary literature

EXAMPLES

- Literature review articles
- Books

Since most information sources in the secondary literature contain extensive bibliographies, they can be useful for finding more information on a topic

TERTIARY LITERATURE

- Summaries or condensed versions of materials
- Usually with references to primary or secondary sources
- Good place to look up facts or get a general overview of a subject

EXAMPLES

- Textbooks
- Dictionaries
- Encyclopedias
- Handbook

Primary Sources of Literature

- 1. Primary Sources: first-hand accounts of events, practices, or conditions being researched.

JOURNAL OF EXPERIMENTAL BIOMATERIALS RESEARCH ARTICLE
JULY 2017
VOLUME 45, ISSUE (2)
PAGES 45-52

The potential of spider silk in biomaterials.
K.M. Madey, Ph.D

- It also includes many original, creative works or artifacts.

Caliban's Rebellion (2008)
Image credit: Shehal Joseph at flickr.com. Permission by CC by 2.0

Primary Sources of Literature

Primary Source Examples:

Humanities :

- Creative works (original literature, fiction or non-fiction, poetry, film, performances)
- Music scores handwritten by the composer(s)
- Diaries
- Regalia or original artefacts
- Magazine or newspaper articles
- Autobiographies
- Memoirs
- Oral histories
- Field notes
- Raw data
- Statues and laws

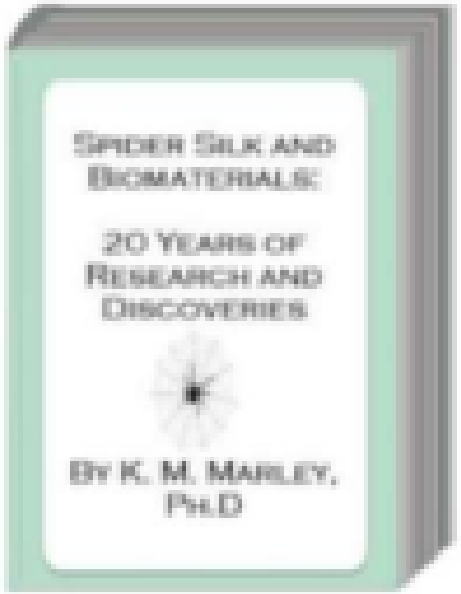
Sciences/Health Sciences :

- Original research (articles, reports, pre-prints).
- Conference papers
- Interviews
- Dissertations & theses
- Patents & invent ions
- Technical reports
- Lab notebooks
- Raw data

Social sciences tend to use a combination from these broad disciplines.

Secondary Sources of Literature

- Secondary Sources: sources that discuss, study, or comment on information from primary sources



THE PERFORMANCE STUDIES PARADIGM

Aurora Szupsel

Abstract: *The aim of this paper is to show how performance can be considered as paradigmatic practice significant for understanding of the contemporary art not only in the filed of theatre and performance art, but also all other artistic and cultural practices as well as a number of non-artistic practices. I will develop my argument based in on the Richard Schechner's understanding of performance as a collection of interpretative tools for studying broad spectrum of social phenomena, where the objects of research don't have to be clarified according to the medium, not bound to specific disciplines and Jon Mckenzie's understanding of performance as meta-historic formation of knowledge and power where performance is established as a general paradigm of contemporary society. In the central discussion I will refer to the significance of the concept of liminality.*

Keywords: *performance, liminality, knowledge, power, contemporary art*

Secondary Sources of Literature

Secondary Source Examples:

Humanities:

- Biographies
- Literary criticism
- Monographs written about a topic
- Reviews on creative works
- Music scores that have been published or changed
- Bibliographies
- Law reviews
- Journal articles (criticism and interpretation

Sciences/ Health Sciences :

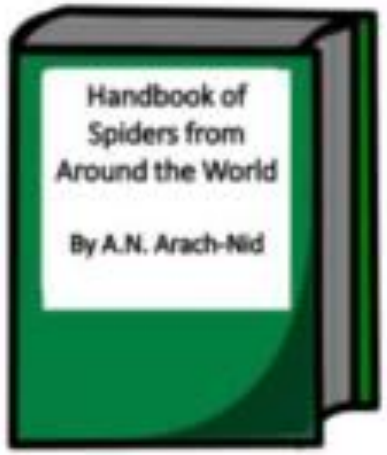
- Literature reviews
- Data analysis or refined compilations
- Reviews (systematic or meta-analysis)Opinion pieces*
- Letters to the editor
- Articles in trade journals (non-research)
- Text books and books

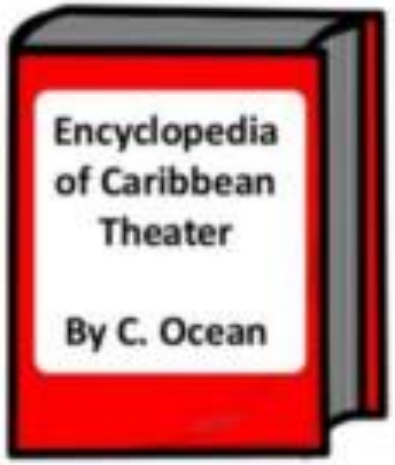
Tertiary Sources of Literature

- Tertiary Sources: sources that utilize and distill information from primary and secondary sources.

Reference works such as :

- Handbooks
- Standards
- Almanacs
- Encyclopedias
- Manuals
- Pathfinders (Lib Guides)





Information and Communication Technology (ICT) plays a crucial role in education, and it has been widely studied in the context of schools and higher education. Here are some key points related to ICT-enabled tools for literature surveys:

- A review of research on ICT integration in education highlights its merits, challenges, and factors influencing successful implementation.
- ICT is considered a powerful tool for educational change and reform, improving educational quality and connecting learning to real-life situations.
- Researchers can use various tools to conduct literature reviews. These tools help organize, analyze, and synthesize existing research.
- Some popular tools include JSTOR, Google Scholar, and PubMed, which provide access to scholarly articles and research papers.

Literature Survey vs. Literature Review

- The Literature Survey only searches information or penetrates through available literature. It is the process of identifying and locating scholarly information and publications on a particular topic.

"Watchingthe skyline."; www.flickr.com

- The Literature Review on the other hand examines the information to determine the findings and contribution to the topic and establish the base upon which any new work would stand. It is the critical review of the literature survey.
- Literature Survey and Literature Review go hand-in-hand simultaneously

Significance of Literature Survey and Review

- It is one of the first things done by any student or scholar who plans to pursue new knowledge or do research in any subject area.
- Time consuming, but helps widen the knowledge base on the topic
- Understand the contribution from other researchers
- Define the gaps of knowledge
- Understand the current status and challenges involved
- Understand advantages and disadvantages of current system
- Trigger the critical thinking process and zero in the research problem to formulate the problem statement

Significance of Literature Survey and Review

This is important to understand the approach and methodology adopted by previous researchers and learn from their experiences.

Some questions that will help in this task are:

- Why you need to pursue research in the selected area?
- How are you going to pursue the research?
- What are the constraints of pursuing the research?
- Which laboratory is to be used for experimentation?
- Is the topic latest or demanding?
- Which is the best method for the experimentation?
- What are the limitations of existing technique?
- Is the research topic feasible?
- Does the problem statement have substantial research component?
- Is there any social use for the selected topic?
- Does it harm the society?
- Is there any parallel research being pursued by another person in the selected area?

Significance of Literature Survey and Review

This will lead to critical information that can be incorporated in the introduction section of documents / thesis such as:

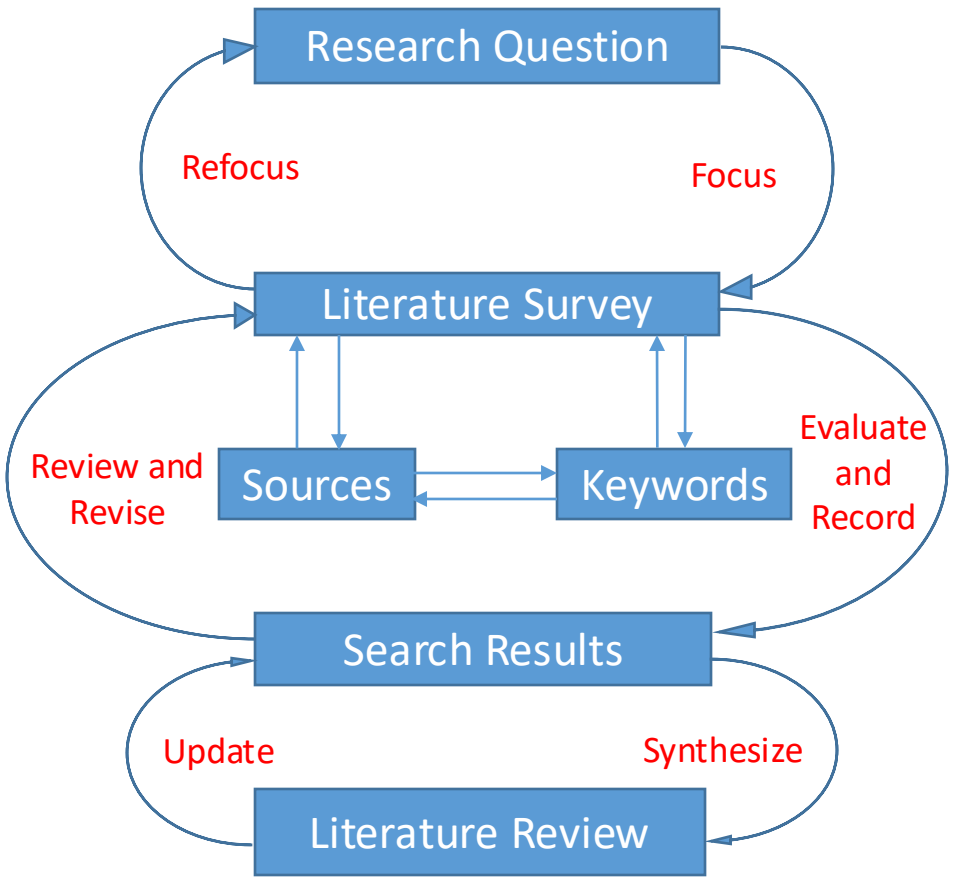
- Introductory detail of the area of interest
- Methodology used for the topic to be studied
- Broad overview of the topic to be studied
- Name of the authors / researchers who are authorities in the area of interest
- Important happenings or events in the area
- Keywords and subject related to the vocabulary terms, which can be used to search the database
- Local, national and international status of the work
- References and bibliographies that can lead to additional search
- Explanation about the fundamental study conducted in the past
- The important historical dates and events of the major work carried out related to the topic
- The information why and in what way the research problem is important

Literature Review Strategy

As literature review is a written summary of journal articles, books, and other documents that describes the past and current state of information on the topic of your research study.

Steps in literature review are:

1. Identify key terms
2. Locate literature
3. Critically evaluate and select the literature
4. Organize the literature
5. Write a literature review
   (Record of research review)



Identify Key Terms (Keywords)

- The words that are typed into a search box are the key to finding the most relevant articles and books. That's why they're called keywords! They need to chosen carefully for effective literature search.
- Write a preliminary "working title" for a project with a brief description.
- Use that to pose a short, general research question (this maybe based on your hypothesis if you already have identified one)
- Use this to identify the keywords in the question and write down the synonyms using a thesaurus.
- Alternatively, the keywords reported by authors in their publications maybe used

Identify Key Terms (Keywords)

Let us take an example:*

- Hypothesis is: "Online classes does not enhance learning by students“
- Research question: "Does the use of online classes help students learn?“
- Key concepts and key words: Online classes, Students and Learn

Probable synonyms that maybe used for searching are:

Students	Online Classes	Learn
learners	online course	Learning
college students	distance education	Success
Undergraduates	zoom	performance
	google meet	grades

Locate Literature

Public and Private Sources

Public sources such as:

- Central government departments - Defense, Energy, Science and Technology etc.
- State and local governments - Highways, Pollution Control Board etc.
- Universities Libraries Internet-free search engines based such as google scholar

Private sources are:

- Non-profit organizations and services - Professional societies, trade associations, membership organizations
- Profit-oriented organizations - Manufacturers, vendors' catalogues, samples, consultants etc.
- Individuals-Direct conversation or correspondence, personal friends, faculty

Locate Literature

Online Databases - Introduction

Online databases:

- Database is a collection of records.
- Computer database is a collection of data presented in a searchable format on computer and some of them are present online.
- Online databases are web-based electronic indexes to locate and retrieve articles, magazines, journals and newspapers.*
- Types of online databases
  - Multidisciplinary (comprehensive) vs. subject specific
  - Full text vs. citation only

Locate Literature

List of Online Databases

- Some important online databases are:
- ERIC database-www.eric.ed.gov - US Dept. of Education
- Psychological abstracts-www.apa.org - Am. Psych. Assoc.
- EBSCO Information services - www.ebsco.com-US Pvt. Co.
- Web of Science - online service by Clarivate Analytics providing access to Science Citation Indexing (SCI), SCI-Expanded (SCIE),Emerging SCI (ESCI), Social SCI (SSCI)
- Sociological Abstracts - ProQuest, USA
- Dissertation Abstracts - ProQuest, USA
- Chemical Abstracts - 1907 to 2009 by American Chemical Society
- Chemical Abstract Service (CAS) - American Chemical Society
- Scopus - Elsevier, Netherlands
- PubMed Central - NIH, USA
- Google Scholar - Google Inc., USA
- Indian Citation Index-ICI, India

Reference Searching and Types

Once a high-impact or key publications on a topic is located, the search can taken further by looking at the references or works cited In those articles. This is called reference searching and is also known by a few different terms: pearl growing, chain searching, and citation mining.

The different types of reference searching are:

1. Systematic-try to find all relevant materials using keywords. This is discussed in detail a later section.
2. Retrospective - find the most recent material and work backwards (Backward Searching or Snowball Method).
3. Citation-follow up references from useful articles, book and reading lists (Forwarding Searching).
4. Targeted-restrict your topic and focus on a narrow area of the literature

The result produced at the end of a literature search is usually a list of references

Reference Searching

Retrospective Type - Backward Searching

- Backward reference searching, also known as chain searching or snowball method, involves identifying and examining the references or works cited in an article. It is done for the following reasons:
  - Learn about the development of knowledge on a topic.
  - Study the origins and development of a theory, construct, or model of interest.
  - Identify experts, institutions or organizations that specialize in a topic of research.

REFERENCES

Aceituno P. 1988. On the functioning of the Southern Oscillation in the South American sector. Part I: surface climate. *Monthly Weather Review* **116**: 505–524.

Alpert L. 1945. The inter-tropical convergence zone of the eastern Pacific region. I. *Bulletin of the American Meteorological Society* **26**: 426–432.

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Barnston A, Glantz MH, He Y. 1999. Predictive skill of statistical and dynamical climate models in SST forecasts during the 1997–98 El Niño episode and the 1998 La Niña onset. *Bulletin of the American Meteorological Society* **80**: 217–243.

Caviedes CN, Knapp G. 1995. *South America*. Prentice Hall: Englewood Cliffs, NJ; 330 pp.

Gray WM, Sheaffer JD. 1991. El Niño and QBO influences on tropical cyclone activity. In *Teleconnections Linking Worldwide Climate Anomalies*, Glantz MH, Katz RW, Nicholls N (eds). Cambridge University Press: Cambridge, 256–284.

Reference Searching

Retrospective Type - Backward Searching

- A second-level or second-generation backward reference search is when a researcher examines sources cited by the references used in an initial article. This allows a researcher to identify inconsistencies in the literature.

- Third-level or third-generation searches are where the references of articles located in a second-level or second- generation reference search are examined. These searches can also go further back.
- Backward Author Searching. The works of an author can be examined by doing a backward reference search. This allows a researcher to review an author's previous work and publications, and also the development of their research

Reference Searching
Citation Type-Forward Searching

- Forward reference searching is when a researcher identifies articles that cite an original article or work after it had been published. This type of search focuses on the publications created after an article's publication.

- Forward reference searching helps a researcher do the following:
  - Expand their knowledge on a topic by locating follow-up studies.
  - Identify new findings and developments on a topic
  - Examine and consider other methodologies or analyses used to study a topic.
- Forward Author Searching. Forward reference searching can also be done to find a particular author and his or her works. This allows a researcher to review an author's later or more current works, and also examine new developments

Review Articles - Definition

- A review article is a:
  - A critical, constructive analysis of the literature in a specific field through summary, classification, analysis, comparison.
  - A scientific text relying on previously published literature or data. New data from the author's experiments are not presented (with exceptions: some reviews contain new data).
  - A stand-alone publication. Literature reviews as integral parts of master theses, doctoral theses or grant proposals will not be considered here. However, many tips in this guideline are transferable to these text types.

Review Articles - Function and Audience

The function of a review article is to:

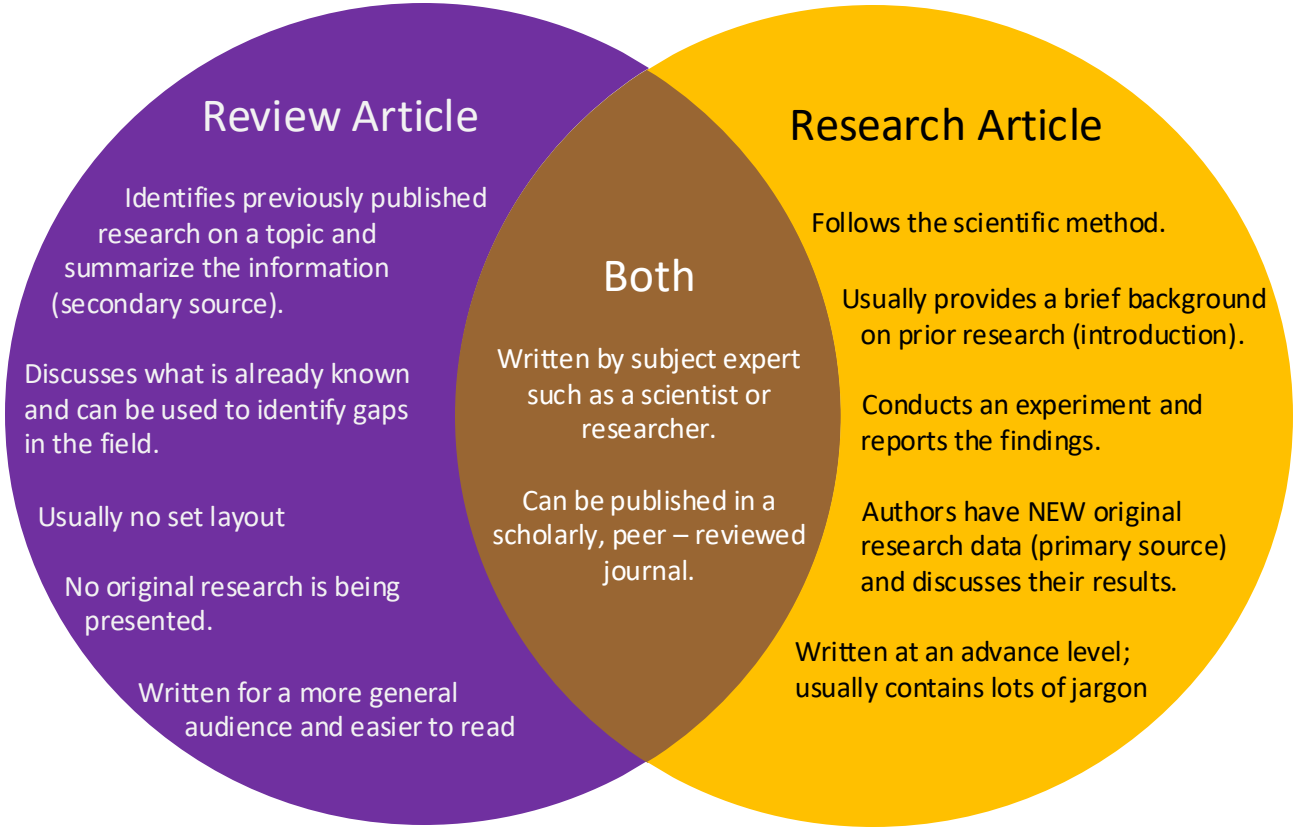
- Organize and evaluate literature
- Identify landmark events in the field
- Summarize the current state of understanding on a topic
- Identify patterns, debates and future trends
- Identify gaps and recommend new research areas

The audience of review articles are:

- Experts in specific research areas
- Students or novice researchers
- Decision-makers

Review Article *vs.* Research Article

- **Research articles** are primary literature sources. So, they report the methods and results of original studies performed by the authors.
- **Review articles** are secondary sources. It is a survey and attempt to provide an overview of the current state of understanding on topics.



Classification of Review Articles

Review articles maybe classified into three types - Methodological, Objective and Mandate based types.

1. Methodological Approach and it types

- Narrative Review:
  - Selected studies are compared and summarized on the basis of the author's experience, existing theories and models.
  - Results are based on a qualitative rather than a quantitative level.
- Best Evidence Review
  - A focus on selected studies is combined with systematic methods of study-selection and result exploration.
- Systematic Review :
  - Findings from various individual studies are analysed statistically by strict procedures.
  - Meta-Analyses are used to pool the results of individual studies

Classification of Review Articles

Review articles maybe classified into three types Methodological, Objective and Mandate based types

- Objective Approach and its types
- Status-quo Review
  - Presentation of the most current research for a given topic or field of research.
- History Review
  - Development of a field of research over time.
- Issue Review
  - Investigation of an issue (i.e. a point of disagreement or a question) in a specific field of research.
- Theory/model Review
  - Introduction of a new theory or model in a specific field of research

Classification of Review Articles

Review articles maybe classified into three types - Methodological, Objective and Mandate based types.

- Mandate Approach and its types:
  - Invited Reviews:
    - Experienced researchers are invited by the journal editors to write reviews when the editors feels that the research has progressed enough in the topic to overview the topic again.
  - Commissioned Reviews:
    - Formal contracts of authors with clients to write and submit reviews
  - Unsolicited Review Submissions:
    - Individual or group of researchers develop an idea for a review and submit it to journal editors

Classification of Literature Review
Another Approach

- Topical order - organize by main topics or issues
- Chronological order - organize by dates the research was published
- Problem-cause-solution order - organize by moves from problem to the solution
- General-to-specific order (Funnel approach)-examine broad-based research first and then focus on specific studies that relate to the topic
- Specific-to-general order - discuss specific research studies so conclusions can be drawn

Review Article - Organization

1. Title - Helping readers to decide whether they should read the text or not
2. List of Authors - Declare intellectual ownership of the work and provide contact information
3. Abstract - Informs about the main objectives and result of the review article
4. Table of Contents - Shows the readers the organization of the text. Helps orientation among sections.
5. Introduction - provides information about the
   - Context (one or two paragraphs)
   - Motivation for review (one paragraph)
   - Focus (one paragraph). Research question (one paragraph) and
   - Text structure (one paragraph)

Review Article - Organization

6. Body-Materials and Methods

- Systematic and best evidence reviews have a methods section.
- Narrative reviews do not have a methods section but should include some information about applied methods at the end of the introduction.

7. Body-Main Review - A coherent structuring of the topic is necessary to develop the section structure. Subheadings reflect the organization of the topic and indicate the content of the various sections. Possible criteria for structuring the topic are:

- Methodological approaches
- Models or theories
- Extent of support for a given thesis
- Studies that agree with another versus studies that disagree
- Chronological order
- Geographical location

Review Article - Organization

8. Conclusion –

- Answer the research question set in the introduction.
- States the implications of the findings and an identifies possible new research field.9.

9. Acknowledgements - Expresses gratitude to

- People who helped with the literature review, but whose contribution is too small to justify co-authorship.
- Funding organisation and specifies the funding program (often required by funding agencies).

10. References ("Literature Review") - organised by number in the order they were cited in the text.

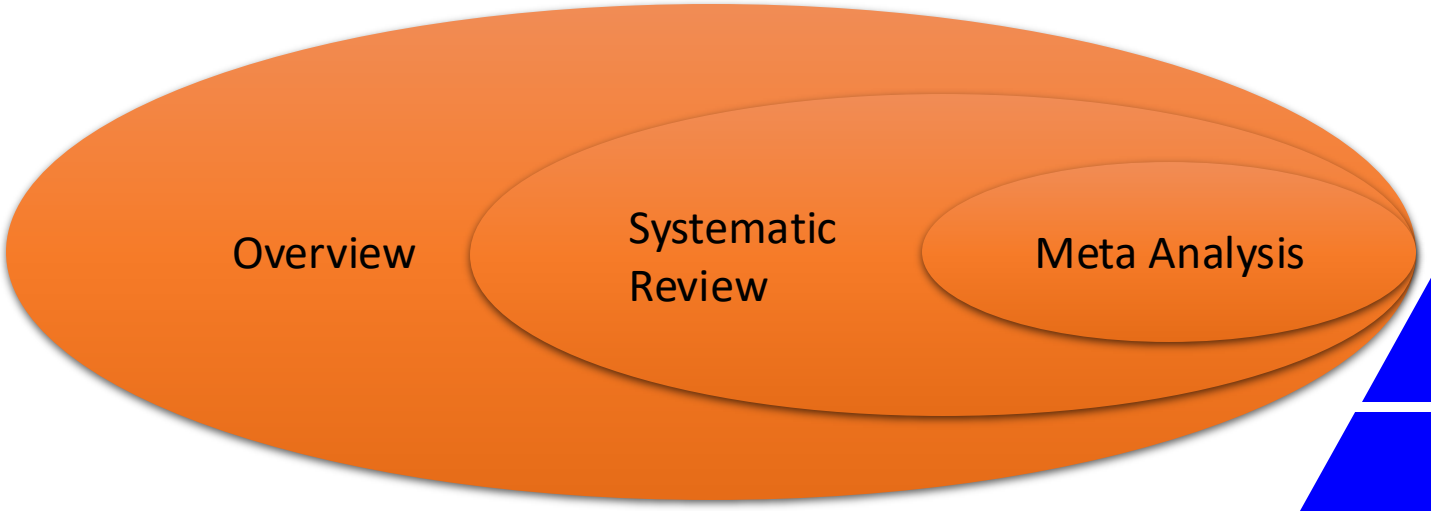
- Shows interested readers how to find the literature mentioned in the text.
- Acknowledges the work of other scientists.
- Compulsory to avoid charges of plagiarism

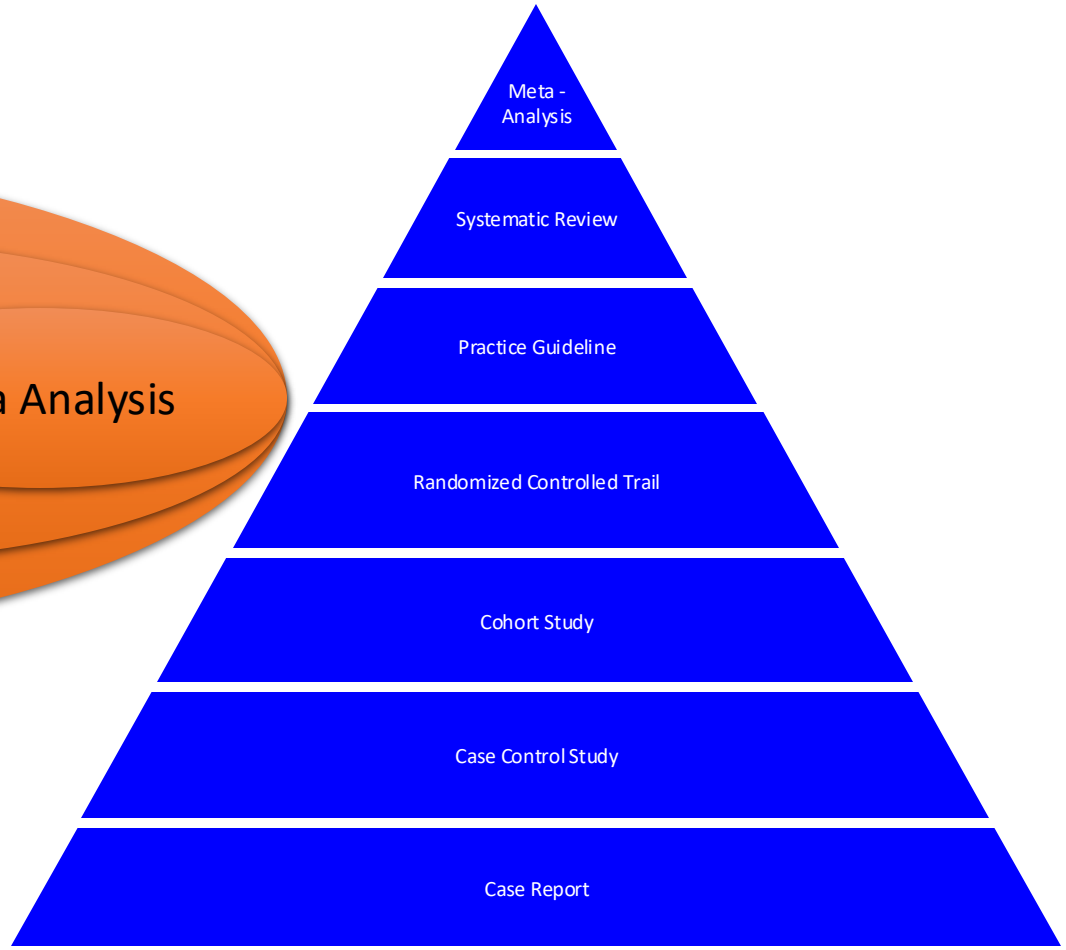
Meta Analysis: "Analysis of an Analysis“

- Coined by Gene V. Glass in educational research
  - Meta-analysis is quantitative approach for systematically combining results of previous research to arrive at conclusions about the body of research.
  - Quantitative : Numbers
  - Systematic : Methodical
  - Combining : Putting together
  - Previous Research : What's already done
  - Conclusions : New knowledge (stronger than the analysis of any single study)
- Meta-analysis is a set of statistical techniques for combining data from independent studies that share a hypotheses to produce a single estimate of an effect.

Meta Analysis: “Analysis of an Analysis”

Meta-analysis is often part of the systematic review process. It is used when you are comparing multiple studies that looked at similar relationships.





Advantages of Meta - Analysis

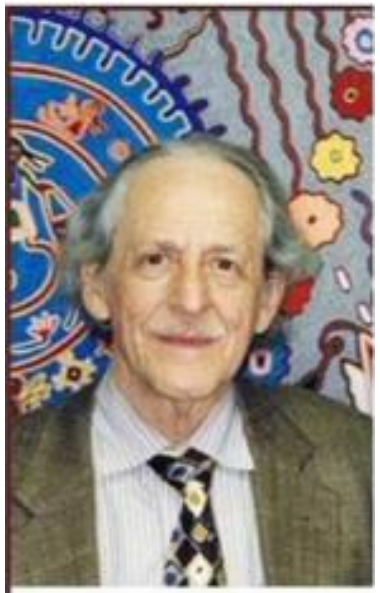
- Results can be generalized to a larger population
- Precision and accuracy of estimates can be improved as more data is used
- Inconsistency of results across studies can be quantified and analyzed
- Hypothesis testing can be applied on summary estimates
- Moderators can be included to explain variation between studies
- Presence of publication bias can be investigated

Research Journal Indexing and Quality Standards

Factors Necessitating Development of Journal Citation Indexing

There are three factors that led to development of citation indexing back in 1950's:

- Followed by World War II, there were huge influx of government funds for research. Publication of research finding by researchers burgeoned the body of knowledge. In order to retrieve from the huge document database, it became necessary to index it. Manual indexing practice is time consuming and not a cost effective one.
- The then method of subject indexing was not useful because of discipline specific terminology etc.
- Availability of machines for computation of data spurred citation indexing
- Citation Indexing proposed and developed by Eugene Garfield



Eugene Garfield

Research Journal Indexing and Quality Standards

- When a manuscript is ready for submission, from the researcher perspective, the criteria used for selecting the journal is the it's subject focus, however, other factors such as good reputation with indexing in major databases also needs to be considered.
- This subject focus + reputation + indexing ensures:
  - Good potential for exposure of the work to research community
  - Greater chance of being discovered and cited by other researchers
  - Greater potential for garnering citations, thereby be able to track the opinion of other researchers and learn how the work contributed to the science and society.
- The reputation of the journal is judged by quality standards as assessed by impact factor and source normalized impact factor.
- From the research journal's perspective, the prestige of the journal is determined by the abstracting and indexing services covering it, which in turn is linked to the citations received by articles published in it

Citation Indices

- The first modern citation index was suggested by information scientist Eugene Garfield in 1955 and created by him in 1964, and it represents an important innovation to knowledge organization and information retrieval.
- Current important citation indices are:
  - Clarivate Analytics citation indices (WoS) - Science Citation Index (SCI); Science Citation Index Expanded (SCIE); Social Sciences Citation Index (SSCI); Arts & Humanities Citation Index (A&HCI); Conference Proceedings Citation Index (CPCI); Book Citation Index (BKCI); Data Citation Index Emerging Source Citation Index (ESCI)
  - CiteSeerx-since 1997, mainly computer and info. science
  - Crossref-since 2000 by Publishers Association using the DOI
  - Scopus Citation Index (Scopus) - since 2004 by Elsevier, largest database
  - Google Scholar (GS)-since 2004 by Google Inc., but poor metrics
  - Microsoft Academic (MA)- since 2016, poor bibliometrics tool
  - Dimensions-since 2018 by Digital Science

Important Citation Indices
Web of Knowledge/Science based by Clarivate Analytics

- Science Citation Index (SCI) and Science Citation Index Expanded (SCIE)- (established 1964, database 1899 to present)
  - A multidisciplinary database, with searchable author abstracts, covering the journal literature of the sciences.
  - It indexes more than 6,650 major journals across 150 scientific disciplines.
- Social Science Citation Index (SSCI) - (established 1973, database 1898 to present)
  - Covering journal literature of the social sciences It indexes more than 1,950 journals spanning 50 disciplines
- Arts & Humanities Citation Index (A&HCI) -(established 1978, database 1975 to present)
  - Covering the journal literature of the arts and humanities
  - It indexes more than 1,395 of the world's leading arts and humanities journals.

Important Citation Indices

- Conference Proceedings Citation Index-Science (CPCI-S) - (established in 2008, database from 1991 to present)
  - Lets you use cited reference searching to see the full impact of conferences and other professional meeting
  - With a global coverage of over 110,000 conference proceedings
- Conference proceedings Citation Index-Arts & Humanities (CPCI-A&H)-(established in 2008, database 1991 to present)
  - This citation index covers conference literature in all fields of social sciences, arts and humanities.

Scopus Database based by Elsevier

- Scopus (established in 2004, database mainly 1996 to present)
  - Largest abstract and citation database of peer-reviewed literature: scientific journals, books and conference proceedings
  - Created by extracting records from child databases and enhanced with their cited references
  - Bibliometric data poor maintained for prior 1996 records

Principles Underlying Journal Impact Factor

- Journal Impact Factor (JIF): Clarivate Analytics (which used to be a part of Thomson Reuters till 2016) publishes JIF in a periodic journal called Journal Citation Reports (JCR) to be used as a quality measure.
- JIF is a part of Web of Science (WoS) database. Clarivate owns and operates a collection of subscription-based services focused largely on analytics which includes Web of Science (WoS).
- JIF is calculated by as the average of the sum of the citations received in a given year to a journal's previous two years of publications (linked to the journal, but not necessarily to specific publications) divided by the sum of "citable" publications in the previous two years.

Computation of Journal Impact Factor

- Consider a journal, SRMIST journal that has been published in 2018, 2019 and 2020.
- Total number of articles published in 2018 = 130
- Total number of articles published in 2019 = 170
- SRM's total number of articles published in 2018 and 2019, say A = 300
- So, SRMIST journal had a total of 300 publications in 2018 and 2019 which had a potential to be cited as a reference by articles that appeared subsequent years, say in 2020 in various journals.
- Please note that there is a possibility that some of the 300 publications are poor quality that would not attract any citations in 2020. Also, some of the articles might be extraordinary enough to attract very high citations in the future years. Hence, different articles will attract different number of citations. All these citations put together regardless of which article attracted how many citations go onto define the magnitude of B.

Computation of Journal Impact Factor

- Let the total number of citations garnered from various journal (SRMIST as well as other journals) in 2020 by the 300 articles that appeared in SRMIST Journal in 2018 and 2019, say B = 600
- Please note the citations in 2020 is not restricted to SRMIST journal and maybe from other journals as well.
- Then JIF for SRMIST journal in 2020 is B/A = 600/300 = 2

$$IF\ of\ Year = \frac{\#\ of\ citations\ to\ Journal\ X\ in\ the\ past\ two\ years}{\#\ of\ total\ citable\ items\ by\ Journal\ X\ in\ the\ past\ two\ years}$$

- Please note that IF for 2020 can be calculated only in 2021 after all journal issues have been published and their citations have been populated in the database.
- The citations garnered in the last two years is a measure of the significance of the journals and is reflected in its impact factor. So, editors try to maximize B value in the expression!

Merits of Journal Impact Factor

- Authors prefer to publish in higher impact factor journals because:
  - Higher citations rate means article has higher chances of getting cited
  - Higher reading rate means article gets read by more researchers
  - More recognition in terms of scholarly contribution
- JIF provides an input about how frequently an average article in a journal has been cited in a particular year
- JIF informs us about the prestige of a journal as a whole
- Editors use it to judge the popularity of their journals and strive to enhance the JIF by choosing good articles that have potential to garner more citations and eliminate ones that might not.
- In libraries, it is a tool used for management of library journal collections (unsubscribe from low JIF ones)
- In market research, publishers use it as a quantitative evidence about the popularity of their journals

Demerits of Journal Impact Factor

- JIF is an average statistic and not all articles are equally cited. So, an article having fewer citations published in a high IF journal is given more importance than a highly cited article in a low IF journal.
- Only includes "citable" articles in the denominator of the equation, i.e., articles and reviews
- Only a limited subset of journals is indexed by Clarivate Analytics
- While calculating JIF, self citations are also counted.
- Tell us nothing concrete about any specific paper or specific author
- Review articles are generally highly cited. Publication of more number of review papers in a journal can greatly affect the journal's IF.
- IF depends on the size of the field concerned. A larger field size would draw more citations than the one having a small size.
- Biased toward English language journals and JCR is also expensive

Journal Immediacy Index

- Journal Immediacy Index (JII) is also reported in JCR and is a measure about how topical or state-of-the-art the topic of a published article might be.
- This is measured by how quickly a published article in a given journal is cited in the same calendar year.
- It is the ratio of total citations for a journal's articles published during a calendar year divided by the total number of articles published in the same calendar year.

Journal Immediacy Index ⓘ

Cites in 2007 to articles published in 2007 = 2343
Number of articles published in 2007 = 366

Calculation: $\frac{\text{Cites to current articles}}{\text{Number of current articles}} = \frac{2343}{366} = \mathbf{6.402}$

- Journals that publish only review articles such as "Chemical Reviews are likely to garner more citations in the same year due to wide applicability of the review articles.

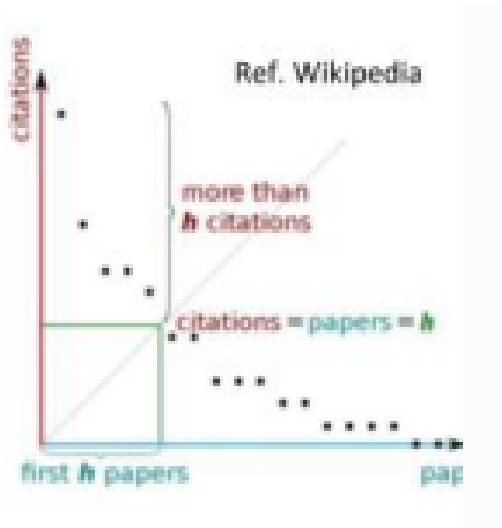
h-index (Hirsch index):

- The h-index was suggested in 2005 by Jorge E. Hirsch, a physicist at UC San Diego, USA and hence also called as "Hirsch Index".
- It is a metric for evaluating the cumulative impact of an author's scholarly output and performance
- It combines an assessment of both quantity (number of papers) and quality (impact, or citations to these papers)
- It is automatically calculated by citation indexing agencies such as Web of Science, Scopus, Google Scholar etc. But, it can also be calculated manually.
- The index can also be applied to the productivity and impact of a scholarly journal as well as a group of scientists, such as a department or university or country

h-index (Hirsch index)

- The h-index is defined as the maximum value of h such that the given author/journal has published "h" papers that have each been cited at least "h" times.
- For example consider the tabulation for "rank of article" vs. "decreasing number of citations for a particular author.
- The articles are listed in decreasing order of number of citations and then plotted for serial number of article vs. number of citations as shown in the plot.
- A diagonal line is drawn across the plot that shows that the h-index is 5. This can also be inferred from the table, where the article serial number and number of citations match

Articles	Citations
1	46
2	25
3	12
4	11
5	5 = H - Index
6	4
7	2
8	1



Factors influencing h-index

- The h-index of a particular author may vary from one database to another, depending on the coverage of each database.
- Co-publishing can elevate the h-index, because it does not take the order of authorship into account.
- Time helps to increase the h-index; young researchers generally2have a low h-index.
- The subject area can have an effect on the h-Index: a researcher who publishes on a trendy subject will have a higher h-index than another one who publishes on a niche topic.
- The language affects h-index; with English receiving more citations.
- Open access publishing increases the visibility of articles and increase their chance to be cited.
- Promotion of publications (blog posts, conferences etc.) helps to make articles more visible and increases their chance of being cited.
- h-index does not eliminate self-citations!

Source Normalized Impact per Paper (SNIP)

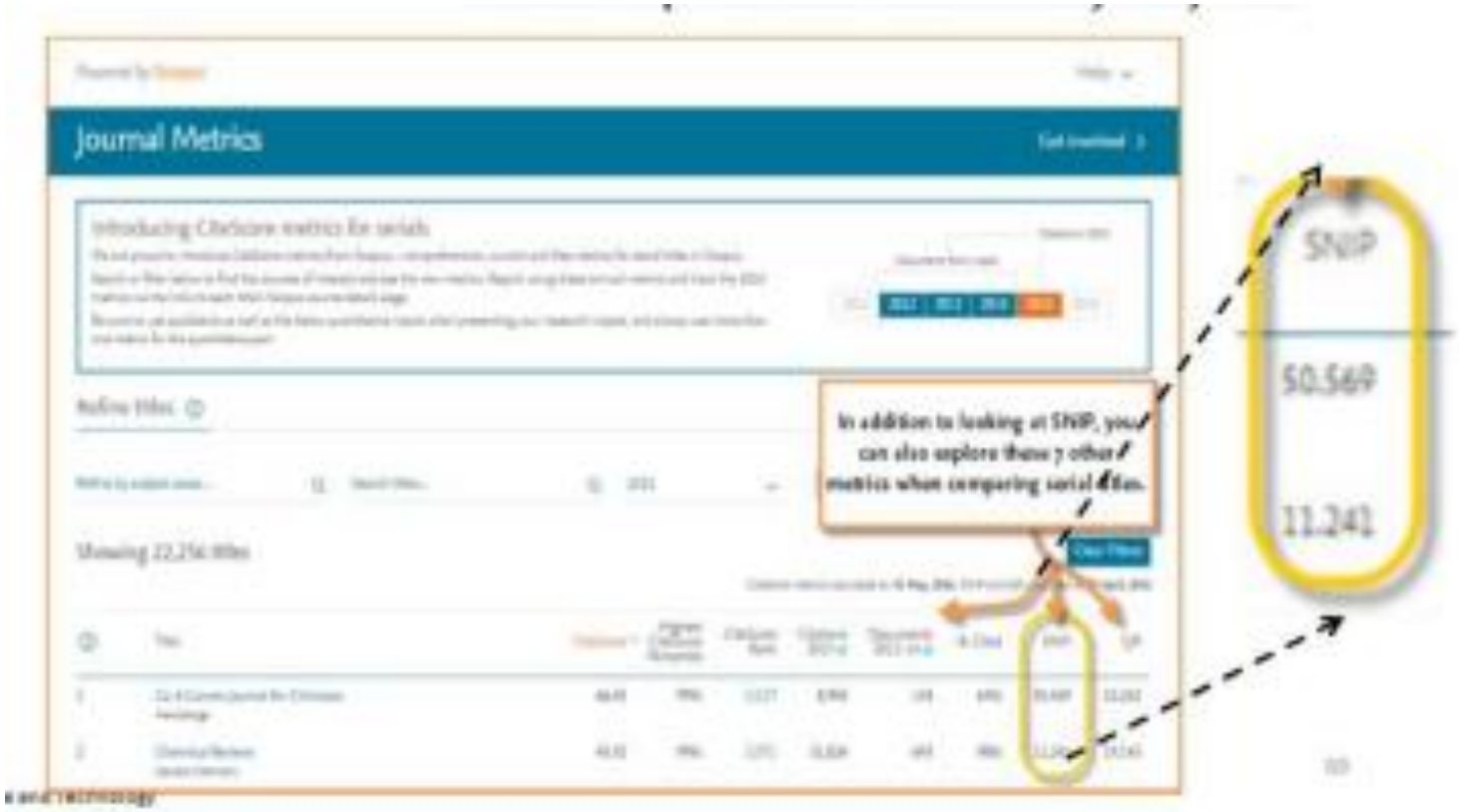
- SNIP is a factor released in 2012 by Elsevier based on Scopus to estimate impact of a journal article.
- The measure is calculated as SNIP RIP/(R/M), where RIP = Raw Impact per Paper, R = Citation Potential and M = Median database citation potential.
- SNIP corrects for differences in citation practices between scientific fields, thereby allowing for more accurate between-field comparisons of citation impact.
- The main differences between the SNIP and JIF are:

Differences between SNIP and JIF		
	SNIP	JIF
Database based on	Scopus	Web of Science
Field Differences taken into account	Corrected	No correction
No. of Year of Cited Publications	Three	Two
Citation Sources	Selected sources only	All sources
Citations Referred to	Selected documents only	All types of documents

Calculation of SNIP

- SNIP can be obtained from www.journalindicators.com.it can also be obtained from the Scopus' Journal Metric webpage as shown below. It can also be calculated as shown below.

$$SNIP\ of\ journal\ = \frac{Journal's\ citation\ count\ per\ article}{Citation\ potential\ in\ the\ subject\ field}$$



i10-Index - Introduction and Computation

- The i10-index was created by Google Scholar in 2011.
- The i10-Index for a particular author is the number of publications with at least 10 citations.
- It is accessible only from Google Scholar Citations profile of the author.
- i10-index is a very simple measure proposed by Google Scholar, and is another way to help gauge the productivity of a scholar.
- Therefore, if someone has an i10-index of 32, it means that the person has 32 papers with citations more than 10. In reality, many papers may have much more than 10 citations (say 100 or even more), but reference frame is at least 10

i10-Index-Introduction and Computation

- Advantages of i10-IndexSerial
  - ➢Very simple and straight forward to calculate
  - ➢My Citations in Google65Scholar is free and easy to use

- Disadvantages of i10-Index
  - ➢Used only in Google Scholar
  - ➢Minimum requirement is 10 citations per article, so there is no weightage for articles with more than 10 citations.

Serial No.	Number of Citation
1	75
2	68
3	65
4	60
5	45
6	15
7	12
8	9
9	5
10	0

i10 Index is 7

H Index is 8

Intellectual Property Rights (IPR)
Definition and Motivation for IPR

- Intellectual property rights are legal rights that provide creators protection for original works, inventions, or the appearance of products, artistic works, scientific developments etc.
- This incentivizes discovery and creativity by providing creators with an opportunity to profit from the value of their innovative work
- The IP rights are important because they can:
  - Used to differentiate a business from its competitors
  - Be sold or licensed providing an important revenue stream
  - Offer customers something innovative that is not available with other businesses etc.
  - This encourages innovation, competitiveness, knowledge economy development, jobs and economic growth and prosperity

Broad Classification of IPR

- Intellectual property is broadly categorized into two:
  i. Industrial Property-Patents, trademarks and designs-eg. drug molecule patent, coca cola trademark and bottle design

  ii. Literary and Artistic Property-copyright and related rights- eg. Shakespeare's literary works, Computer codes, Video recordings, Musical works, Website design, Pictures etc

World Trade Organization

- The World Trade Organization (WTO) is a global international organization dealing with the rules of trade between nations.
- It is based on WTO agreements signed by the bulk of the world's trading nations.
- Its goal is to ensure that trade flows smoothly, predictably and as freely as possible between nations.
- It acts as a forum for negotiating trade agreements, settling trade disputes between its members and supporting the needs of developing countries.
- India is a member of WTO since inception in 1995

World Intellectual Property Organization

- The World Intellectual Property Organization (WIPO) was established in 1967 and is a specialized agency of the United Nations with 193 member states.
- WIPO provides a platform with treaties, forums, standards, databases and services to register and protect IP in different countries, resolving transboundary IP disputes etc. for a balanced and accessible international IP system which will rewards creativity, stimulates innovation and contributes to economic development while safeguarding public interest.
- India is a member to several WIPO conventions and treaties, thereby underlining its strong multilateral engagement with WIPO and intent to create a strong regime for IP protection

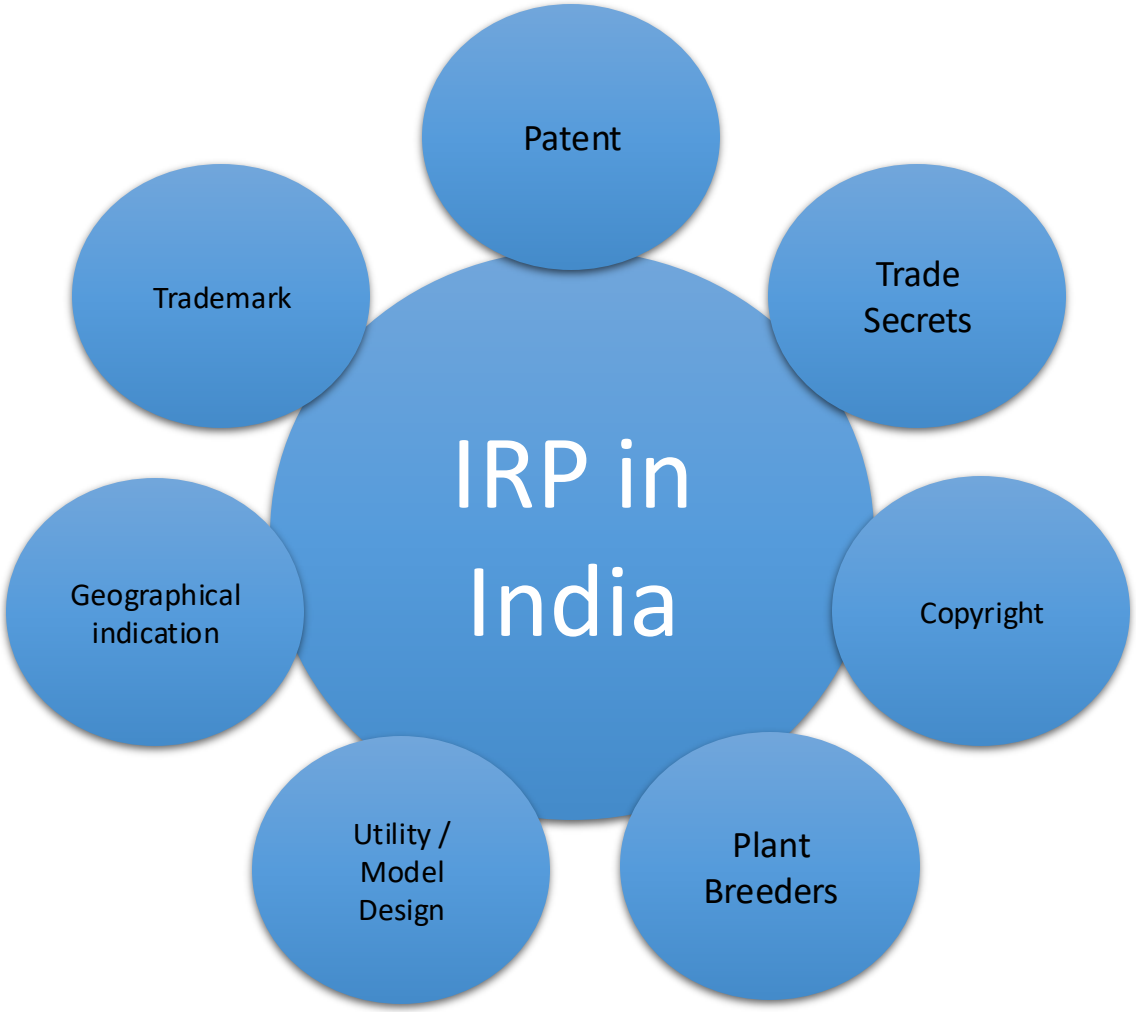
Trade-Related Aspects of Intellectual Property Rights Agreement

- The widespread piracy, counterfeiting and IPR infringement constituted a major barrier to trade between nations, inspite of the WTO agreement.
- To reduce these impediments, resolve IP related disputes and support a mutually supportive relationship between WTO and WIPO, the Trade-Related Aspects of Intellectual Property Rights (TRIPS) agreement was constituted in 1995.
- Hence, TRIPS is an international legal agreement between all WTO member nations.
- It establishes minimum standards for resolution of IP related trade disputes between WTO members while taking into account the domestic interests of the nations.
- The Agreement is a legal recognition of the significance of links between IP and trade and the need for a balanced IP system

Types of Indian IPR

IPR in India can be classified as follows:

1. Patents (and inventions)
2. Design (Industrial Design)
3. Trademark
4. Copyright
5. Geographical Indication
6. Trade Secret



Types of Indian IPR

1. Indian Patents :

- A patent is used to prevent an invention from being created, sold, or used by another party without permission.
- Patents are the most common type of intellectual property rights that come to people's minds when they think of IPR protection.
- A patent owner has every right to commercialize the patent, including buying and selling the patent or granting a license to the invention to any third party under mutually agreed terms.
- A patent has a 20 year validity under Indian Patent Act, 1970.

Indian Patent Certificate

Factors to be Considered in Patenting

- A patent is a "negative right" - it gives the right to EXCLUDE others from practicing your invention
- All rights in patents (derive from the inventors)
- Patentable vs. Non-patentable
  - Patentable (utility, plant and design) - Ie. process, machine, manufacture, or composition of matter, "everything under the sun that is made by man“
  - Non-patentable-abstract ideas, laws of nature, and natural phenomena

Factors to be Considered in Patenting

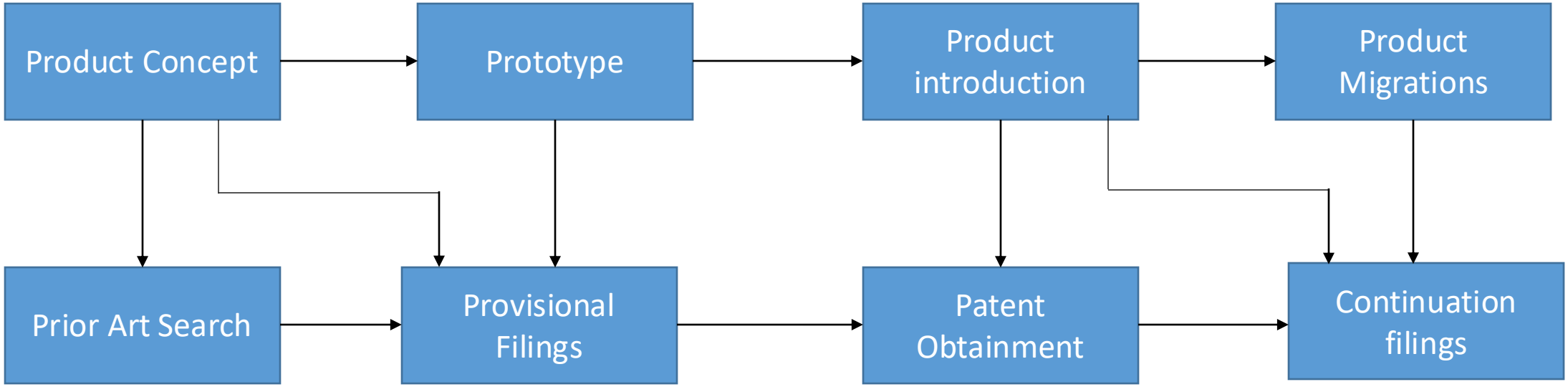
- Obtaining a patent can be expensive and arduous process
  - Average time from filing to first response from PTO: 1.5-3 years
  - Average time to obtain patent: 3-5 years
  - Average cost of U.S. patent: US $ 20,000-40,000
  - Can file "continuations" (P-> C1-> C2 -> CIP)
- Patent litigation is the most intellectually challenging type of IP litigation
- Ultimate aim of patent is to promote public good by providing royalties for inventors & companies which invest in new inventions
  - Royalty rate per "use" is the monopoly price
  - Marginal cost=0, so revenue maximization

Procedures for Obtaining Patent Rights

- Step 01: Invention disclosure.... with novelty and disclosure to agent if necessary
- Step 02: Patentability search.... for prior art by agent
- Step 03: Decision to file an application for patent..... with a inventive step
- Step 04: Patent drafting.... with scope and detail
- Step 05: Filing the patent application....
- Step 06: Request for examination.... by the officers
- Step 07: Responding to objections (if any)...
- Step 08: Grant of patent.....
- Step 09: Renewal of patent...

Intellectual Property Rights (IPR)
Product Development as a Case Study

- Property right-control over usage
- Each right protects a different feature or aspects
- Various aspects of work protected by one or more of these rights



Copyright Laws vs. Patent Rights

Copyright	Patent Rights
Protects expression, not ideas	Protects ideas that have been reduced to practice
Protects the use or performance of original works of literature, art, music, drama, or any other form of expression	Protects the use manufacture or sale of inventions
Life of author + 70	20 years from filing
Fair Use	No Fair Use, No Research Exemption

Copyright Laws vs. Patent Rights

Copyright	Patent Rights
Works for hire	Employee inventor owns
Protects against copying	Innocent Infringement not a defense
Protection is automatic (registration is relatively simple)	Extensive examination process before any rights granted
Originality (low bars)	Novelty and Nonobvious (high bars)
In software, copyright automatically covers the authorship in the source code	Patentability standard for software patents is strict and uncertain